



CPE 334

**SEMESTER 1/2569 · WEEK 6 ·
LECTURE 6**

Project Management

Phases, goals, and starting a project

INSTRUCTOR

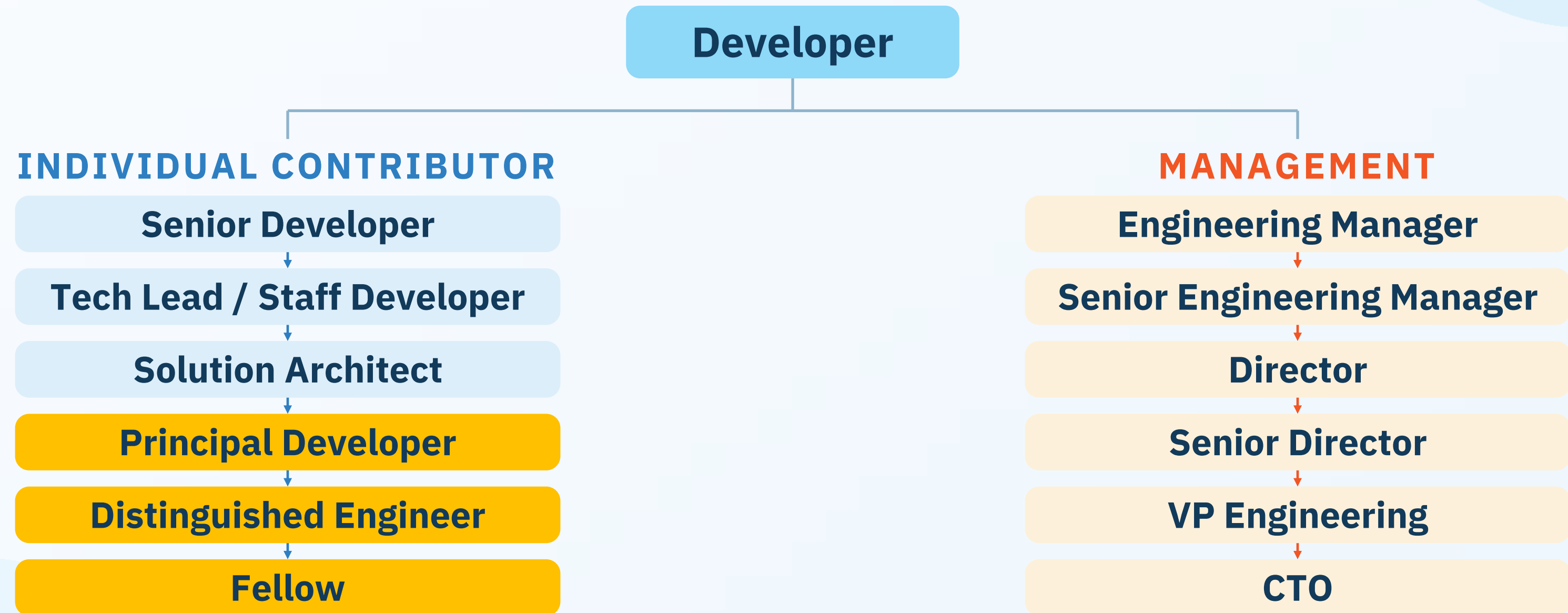
Dr. Piyanit E.
Dr. Suthep M.
Dr. Santawat T.

DEPARTMENT

Computer Engineering, KMUTT

— CAREER PATH

Developer ladders



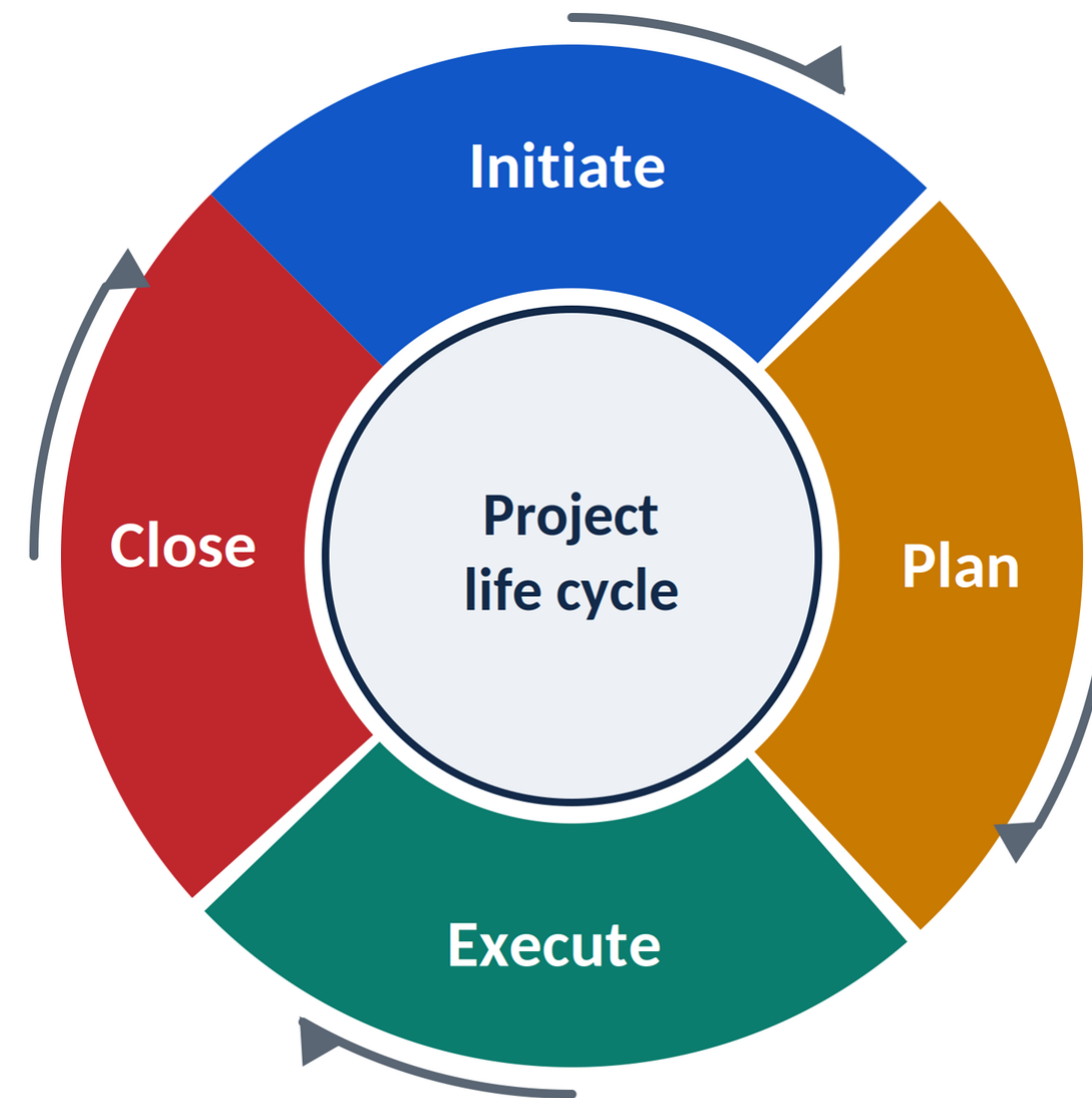
— PART 01

What project management is

— LIFE CYCLE

Project Life cycle

The Project Life Cycle



— TWO LIFE CYCLES

Project Life Cycle vs SDLC

PROJECT LIFE CYCLE

Initiate

Plan

Execute

Close

SOFTWARE DEVELOPMENT LIFE CYCLE

**Requirements &
specification**

Design

Code

Test

Release

**Documenting
& Maintain**

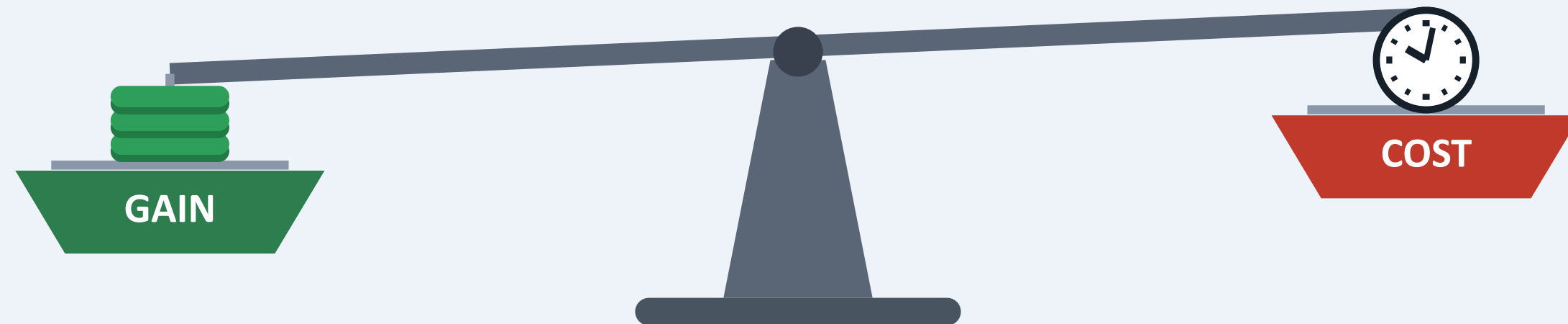
— PART 01

Project initiation

How to start a successful project

Cost–benefit analysis

$$(G - C) \div C = ROI$$



TANGIBLE GAIN

- value created
- money saved
- new revenue
- time saved
- better experience

INTANGIBLE — NEVER ENTERS THE ROI

retention

morale

brand

...or those same three put at risk

TANGIBLE COST

- people's hours
- one-time costs
- ongoing costs
- long-term costs

Only the tangible columns go into the ROI. The intangible panel is considered, not calculated

Working out the ROI

The project costs \$6,000 up front plus \$25 a month for 12 months — \$300 a year — so the total cost is \$6,300.

You estimate it will bring in \$10,000 of revenue over that year.

$$G = \$10,000 \quad C = \$6,300$$

$$(\$10,000 - \$6,300) \div \$6,300 = \text{ROI}$$

$$10,000 - 6,300 = \$3,700 \quad 3,700 \div 6,300 = 0.5873 \quad \times 100 = 58.7 \%$$

A strong ROI tends to be anything above 10 %, so 58.7 % is strong — you pursue the project.

1. ความคุ้มค่าต้อง "เกิน 3.9 เท่า" (High ROI)

หน่วยบริหารจัดการทุนด้านการเพิ่มความสามารถการแข่งขัน ให้ความสำคัญเป็นอย่างมากกับผลลัพธ์ทางเศรษฐกิจ (ยอดขาย หรือ ต้นทุนที่ลดลง) ที่จะเกิดขึ้นในช่วง 5 ปีหลังจบโครงการ โดยผลลัพธ์ที่เกิดขึ้นต้องมีมูลค่าสูงกว่างบประมาณที่ขอรับทุน

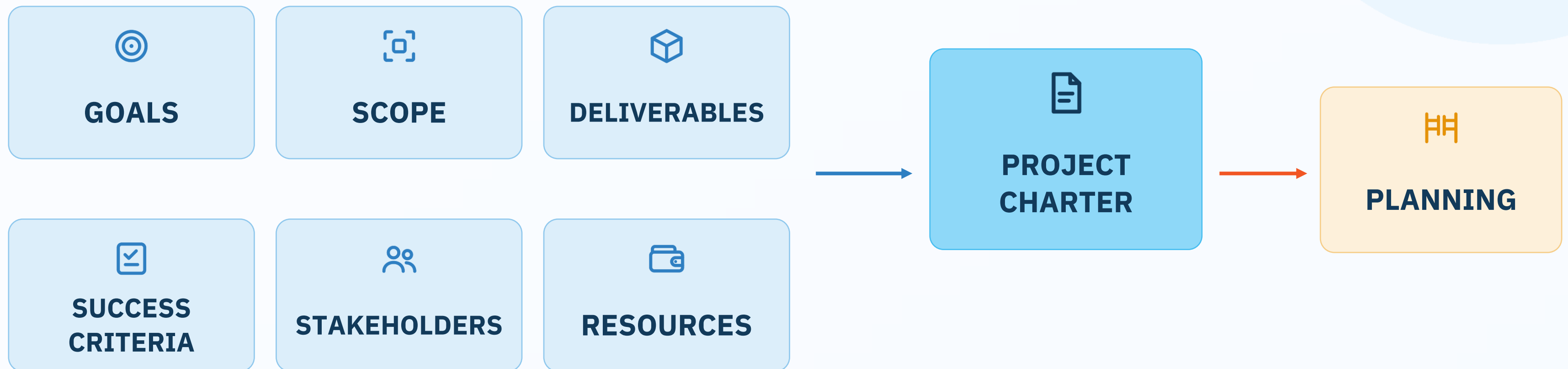
- พิจารณาจากมูลค่าผลลัพธ์ > 3.90 เท่า ของงบประมาณที่ขอ (เช่นของบประมาณ 10 ล้านบาท ต้องสร้างมูลค่า/ยอดขายได้ 39 ล้านบาทขึ้นไป)

Expect gain > 3.90 of Cost
Cost 10 M → expect gain 39 M

$$(39-10) \div 10 \\ = 2.9 \text{ or } 290 \% \text{ ROI}$$

— PART 01

The six components of initiation

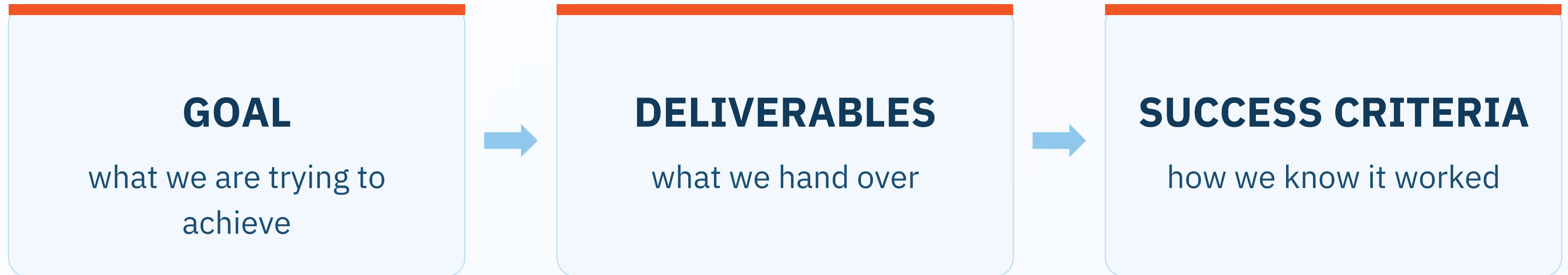


All six are decided during initiation, into the charter, and approved before planning begins.

 PART 03

Goals and deliverables

What to settle before work starts



Each one is defined before the next can be defined.

— PART 03

A well-defined goal

Reduce the median time to first response on support tickets **by 20 per cent, **by** routing them automatically to the assigned queue**

Improve the response time **via email **by 20%****

Increase revenues **through a new service offering **by 5%****

- WHAT you are trying to achieve
- BY HOW MUCH
- HOW you will do it

A well-defined goal is both specific and measurable.

— PART 03

Lab Project Goals

By the end of Lab 3, every protected API and screen is enforced by role and ownership on the server, not by hidden buttons

By the end of Lab 3, IT Staff can find, claim and update work in one shared Ticket Queue with search, filters and sorting

— PART 03

Reviewing goals at the start

-  When a project starts, take time to review its goals and confirm they are well defined
-  You may need more information from your stakeholders to do it
-  Talk to them about their vision for the project
-  Ask how it aligns with the organisation's larger goals and mission
-  By the end of that conversation you and they should agree to support the goals

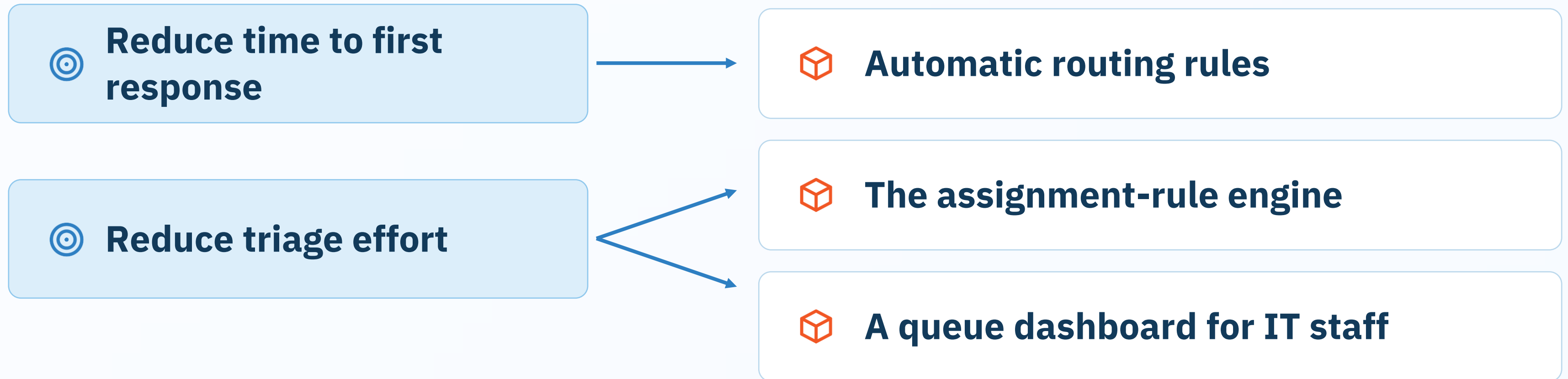
— PART 03

Project deliverables

-  The products or services created for the customer, client or project sponsor
-  What gets produced or presented at the end of a task, event or process
-  Deliverables are how the impact of a project is quantified and made visible

— PART 03

Goals and their deliverables



They are deliverables because they are tangible outputs that show how the reduction will actually be produced.

— PART 03

Defining deliverables with stakeholders



Well defined

Deliverables need to be well defined for the same reasons goals do



Decided up front

Usually settled at the start, with the stakeholders or clients involved



Accountable

They hold everyone accountable, and are a large part of achieving the goal



Ask out loud

Ask what each deliverable should be, and have everyone share their expectations of it

— PART 04

SMART goals

— PART 04

SMART

S



Specific

M



Measurable

A



Attainable

R



Relevant

T



Time-bound



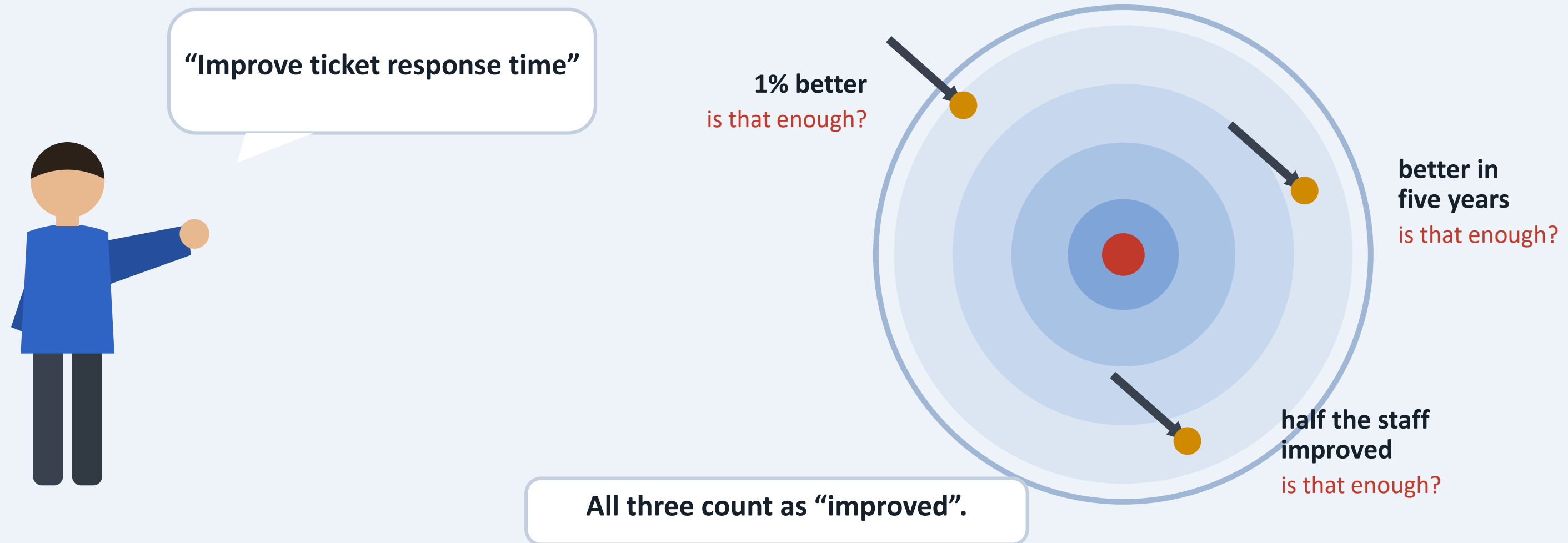
You may not set a project's main goals, but you must be able to identify and clarify them



Deliverables depend on goals, so the goals are worth the effort

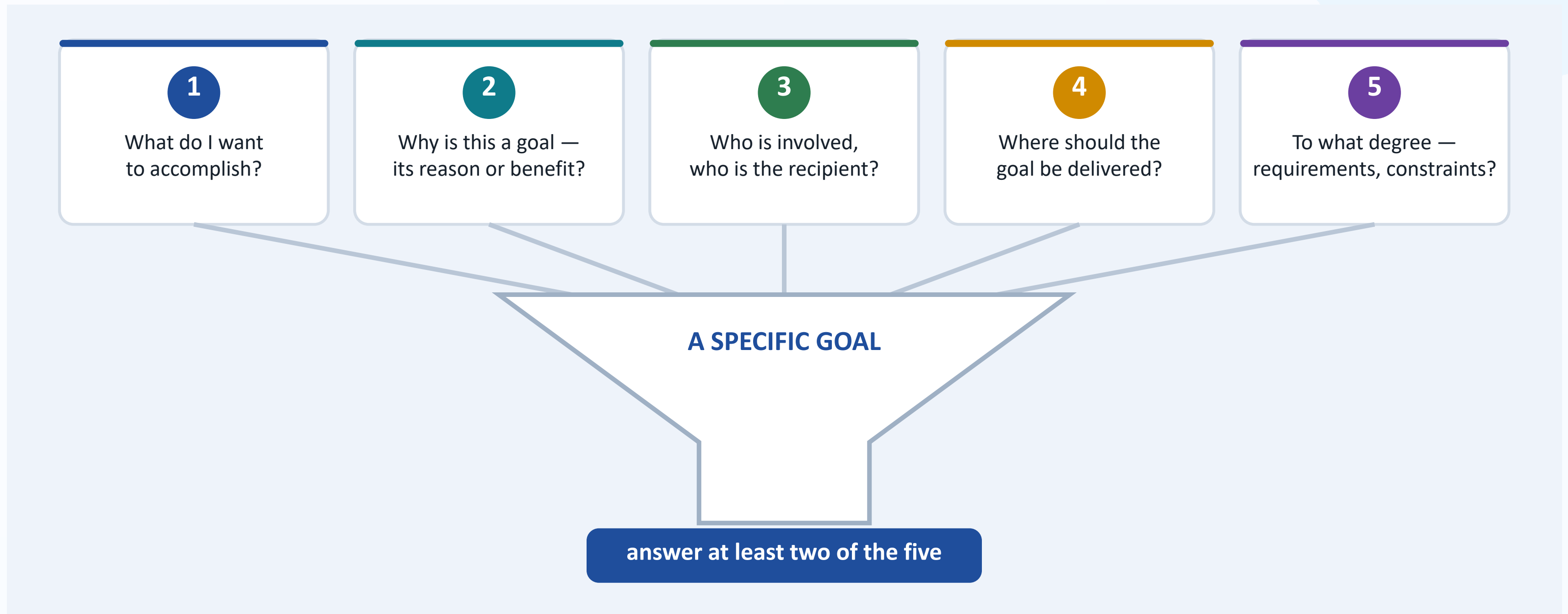
— PART 04

What an unspecific goal leaves open



— PART 04

Five questions for a specific goal



— PART 04

Measurable



Measurable means you can determine that the goal was objectively met



Measuring lets people track progress



It is also what keeps people motivated



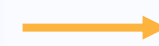
Ask how much, how many, and how you will know when it is accomplished

— PART 04

Metrics

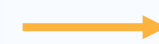
- ✓ Sometimes success is a simple yes or no
- 📊 Most goals need a metric: what you use to measure something, in numbers or figures

Reduce time to first response



median hours to first reply

Reduce manual triage



tickets routed without human intervention

Beware of vanity Metrics! Unless the customers need it

— PART 04

Benchmarks



Use benchmarks, or points of reference, to check the metric you chose is the right one



Last term's figures are the benchmark for deciding what this term's target should be



If response time improved three per cent last term, twenty per cent this term needs a reason behind it

— PART 04

Attainable

Can It reasonably be reached? How can it be accomplished?



Aim a little challenging — if nothing has to change, the goal is not doing any work



Too extreme and you will never reach it: you have failed before you started

— PART 04

Relevant

-  Does it make sense to try to reach this goal
-  How does it line up with other goals, priorities and values
-  Does the effort involved balance out the benefits
-  Does it match the organisation's other needs and priorities
-  Consider the time it will take, and whether the project can be sustained over that time

— PART 04

Time-bound



Time-bound means the goal has a deadline



Deadlines are what let you track progress



Without one you may never reach the goal, or never start



A deadline is also what lets you break the goal into monthly and weekly pieces

— PART 04

All five on one goal

Reduce the median time to first response by 20 per cent this semester, by routing tickets automatically to the assigned queue.

S 🔍 Reduce time to first response, by automatic routing

M 📊 Median hours to first reply

A 📈 About five per cent a month

R 🔗 It is the complaint the IT staff actually make

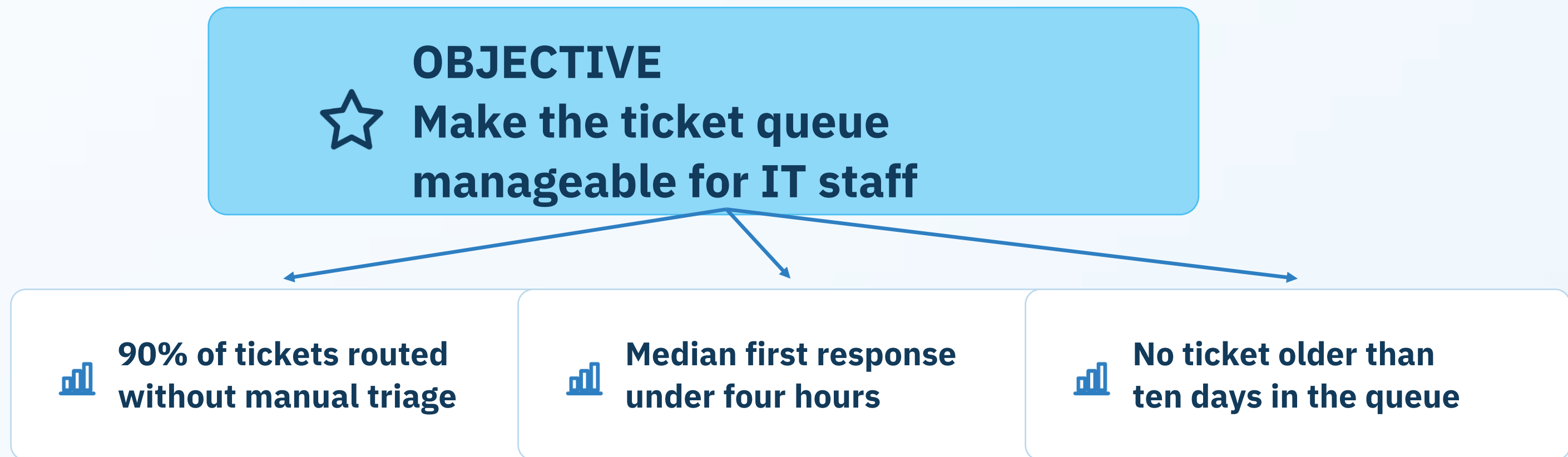
T 🕒 This semester

 PART 05

OKRs

— PART 05

The objective and its key results



KEY RESULTS

The objective says what to achieve. The key results define when it has been achieved.

— PART 05

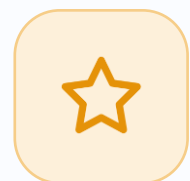
Where OKRs diverge from SMART



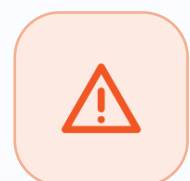
SMART requires a goal to be attainable — practical to achieve



Key results are deliberately more ambitious than that



OKRs are used to set stretch goals: to attempt what has not been done before



Achieving every key result may mean the OKRs were set too easy

— PART 05

Three levels of OKR



ORGANISATION

Shared across the organisation, updated annually, supporting its mission



DEPARTMENT

Supports the organisation's, and drives team performance



PROJECT

Set during initiation, tracked through planning and execution



**ALIGNS
UPWARD**

— PART 05

How a project comes out of an OKR

☆ **Organisation objective**
Reduce the support burden on IT staff



📊 **Organisation key result**
The three most-requested workflow tools in pilot by Q2



🚩
THE TokTickIT PROJECT



Project objective: give IT Staff one shared queue to work from
Key result: every ticket has an owner, claimed from the queue

A company measure becoming somebody's project is where projects actually come from.

— PART 02

Scope

— PART 02

The definition of scope

“An agreed upon understanding as to what is included or excluded from a project.”



Agreed is the load-bearing word — **it is a shared understanding**

— PART 02

What a defined scope knows

AUDIENCE

Who it is delivered to,
and who uses the result

COMPLEXITY

Is it straight-forward? Can it be done with
list of task? Or need a year long research?

THE PROJECT

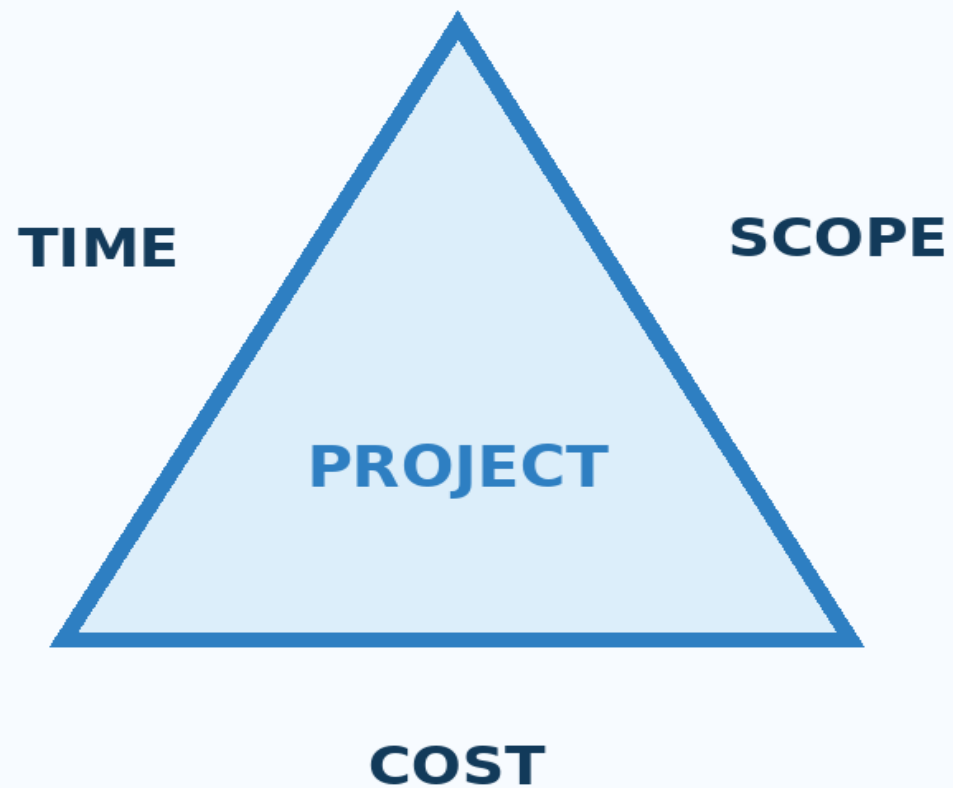
CONSTRAINTS

The timeline, the budget and the resources available

Without all three defined, you cannot say what is actually possible.

THE TRIPLE CONSTRAINT

Scope, time and cost — the three primary variables in project management



SCOPE

what's being delivered — the extent, range and spread of the work to be done on a project

TIME

the duration or amount of time required or allocated to complete the project or its tasks

COST

the resources, financial and otherwise, required for execution — labour, hardware, software, charges

Redrawn from the triple constraint model described in Ben Aston, "Triple Constraints in Project Management: 2026 Guide", The Digital Project Manager, 4 Jul 2022, updated 6 Feb 2026. Figure drawn for CPE 334 — not reproduced from the article.

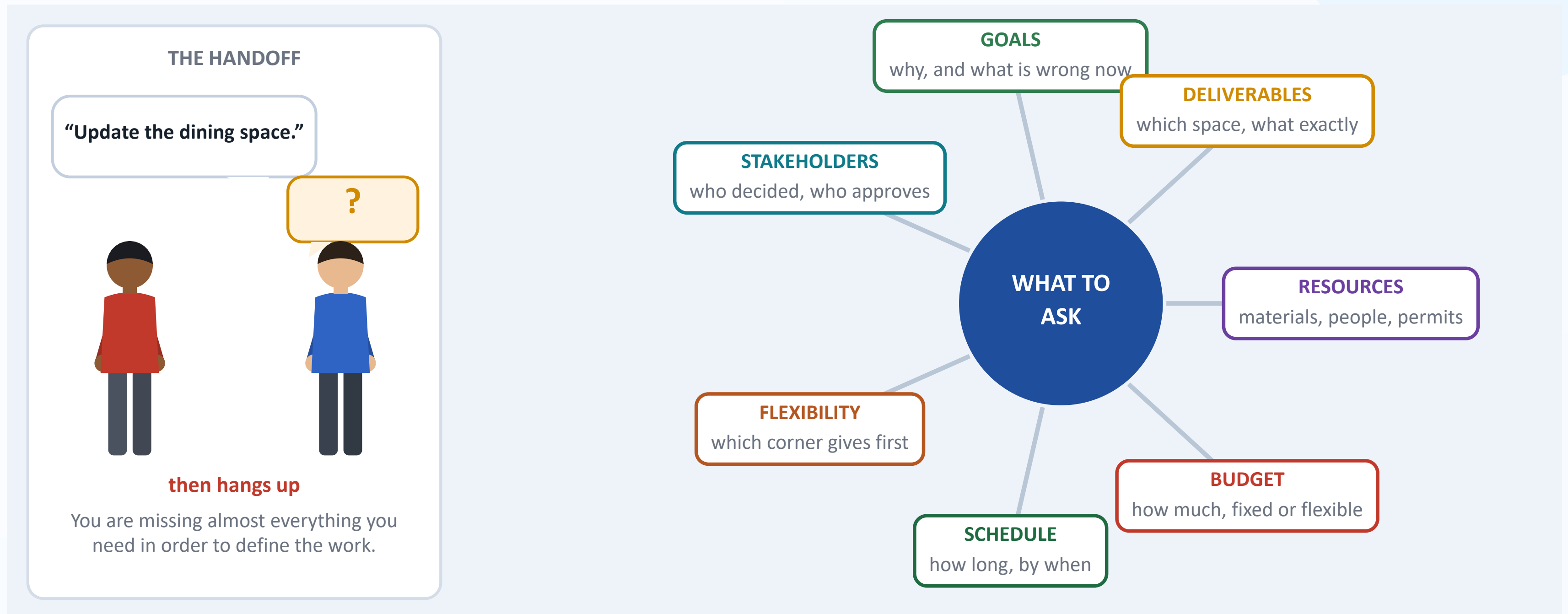
— PART 02

How to establish scope



The exclusions are the half that gets skipped, and the half that causes the arguments.

The questions that establish scope



— PART 02

TocktickIT In-scope and out-of-scope

IN-SCOPE

**Role-based
authorization**

IT Staff ticket queue

**Claim, IT Priority,
status**

Comments and notes

OUT-OF-SCOPE

**Email, SSO, self-
registration**

— PART 02

When scope is defined



Defining scope belongs to the initial planning stage



Start early, so everyone agrees to the same set of expectations



Early agreement is what mitigates the risk of large changes later



Scope can still be adjusted as planning continues, if it needs to be

— PART 06

Scope creep

— PART 02

An out-of-scope request

Tasks that are not included in the project plan and don't contribute to the project's goal



Midway through the build, the team proposes adding email notifications for every ticket state change.



A good idea

Nobody is arguing that it would not help



Out of scope

It is not in what everyone agreed to build



Extra time

It would take development time the plan does not hold



Extra cost

It would add to the cost of the project

ONE SIDE CANNOT MOVE ALONE

“You can’t adjust one side without impacting the others.”



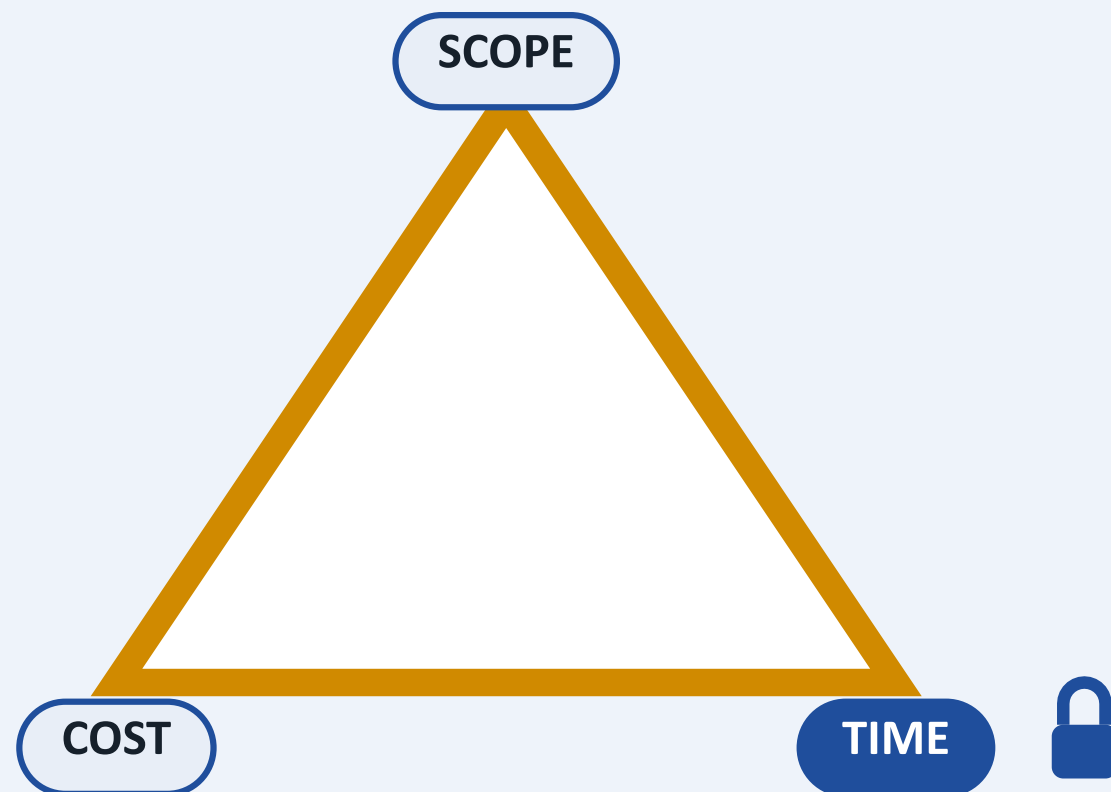
“If you’re going to ask for something to be delivered faster (time), you have to pay more (cost).”
“Projects cannot be simultaneously cheap, good, and fast.”

Redrawn from the triple constraint model described in Ben Aston, “Triple Constraints in Project Management: 2026 Guide”, The Digital Project Manager, 4 Jul 2022, updated 6 Feb 2026. Figure drawn for CPE 334 — not reproduced from the article.

— PART 07

Knowing which corner is fixed

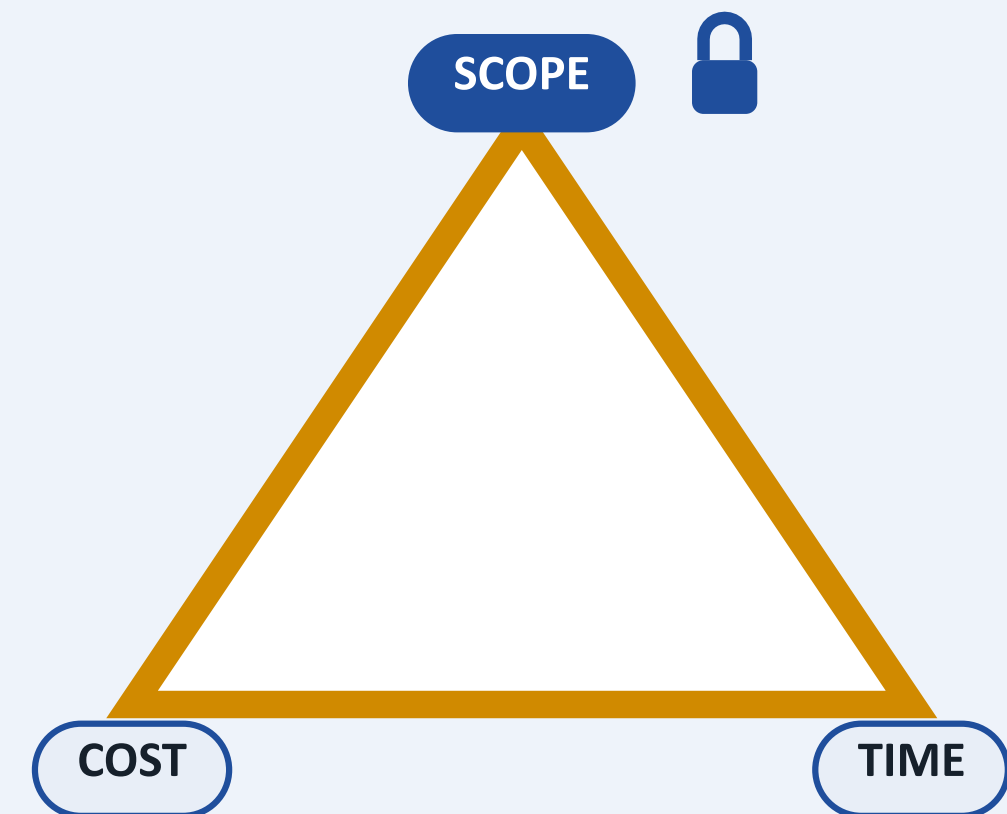
To weigh a trade-off you need to know the project's priorities: which of scope, time and cost matters most.



The deadline is fixed

A hard date is the constraint

→ scope has to give



The product must work a certain way

The requirements are the constraint

→ time or cost has to give

MORE THAN THREE

PMBOK recognises constraints beyond the triangle



Aston's caveat: managing a project is always more complex than the three iron-triangle variables suggest — “probably more like an octagon”. The article names four additional constraints; it does not enumerate a full eight, so neither does this figure.

Redrawn from the triple constraint model described in Ben Aston, “Triple Constraints in Project Management: 2026 Guide”, The Digital Project Manager, 4 Jul 2022, updated 6 Feb 2026. Figure drawn for CPE 334 — not reproduced from the article.

— PART 06

The two sources of scope creep

🌐 EXTERNAL

- A customer requests changes
- The business environment shifts
- The technology changes




**PROJECT
SCOPE**

🏠 INTERNAL

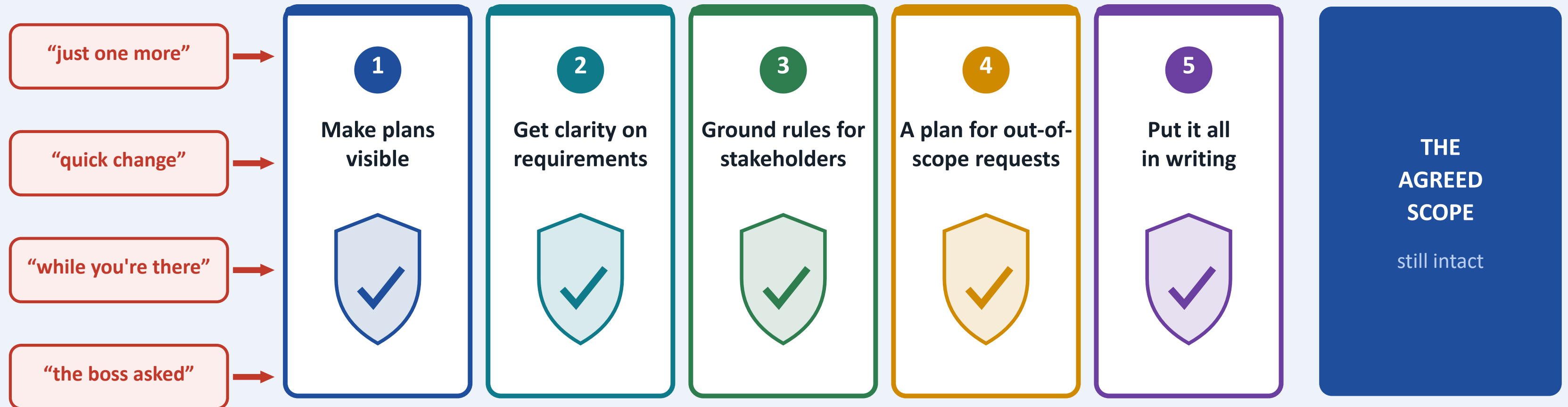
- Team members suggest changes
- Or insist on improvements
- Harder to spot, harder to refuse



You cannot control everything that happens to a project.

Solutions

REQUESTS ARRIVING



— PART 06

The leading cause of external creep



The leading cause is unclear requirements when the scope was defined and approved



This is where specific and measurable goals and deliverables earn their keep



If the requirements are not specific, and the processes, deliverables and milestones are not agreed, scope creep is close to guaranteed

SMART is what stops this happening.

— PART 06

Responding to a change request



Offer alternatives

Another solution, the added risk, a cost-benefit analysis



Learn to say no

Refusing a key stakeholder is uncomfortable and sometimes necessary



Show the interference

Explain how the request cuts into the budget, timeline or resources agreed

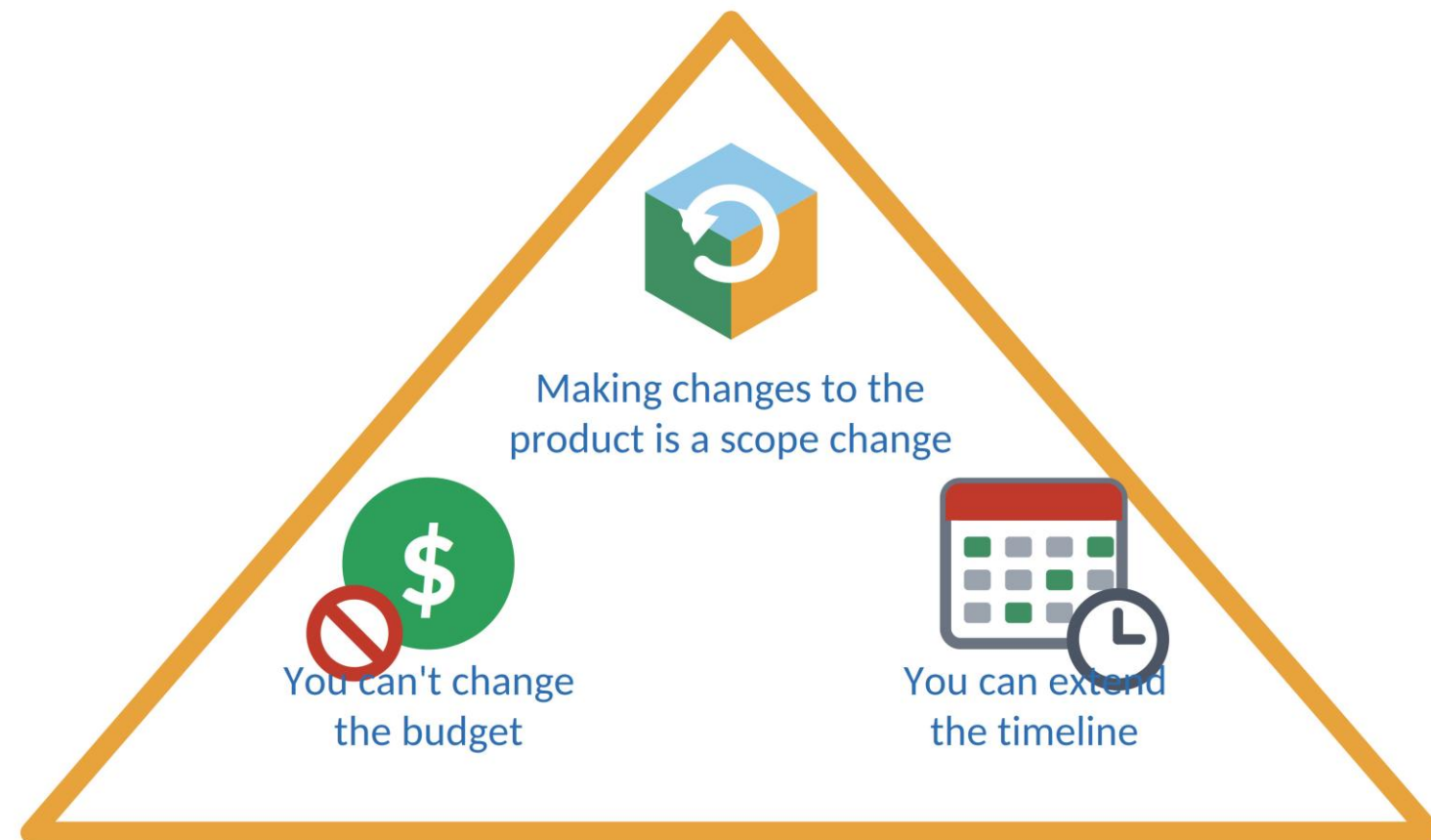


Documents cost of out-of-scope work

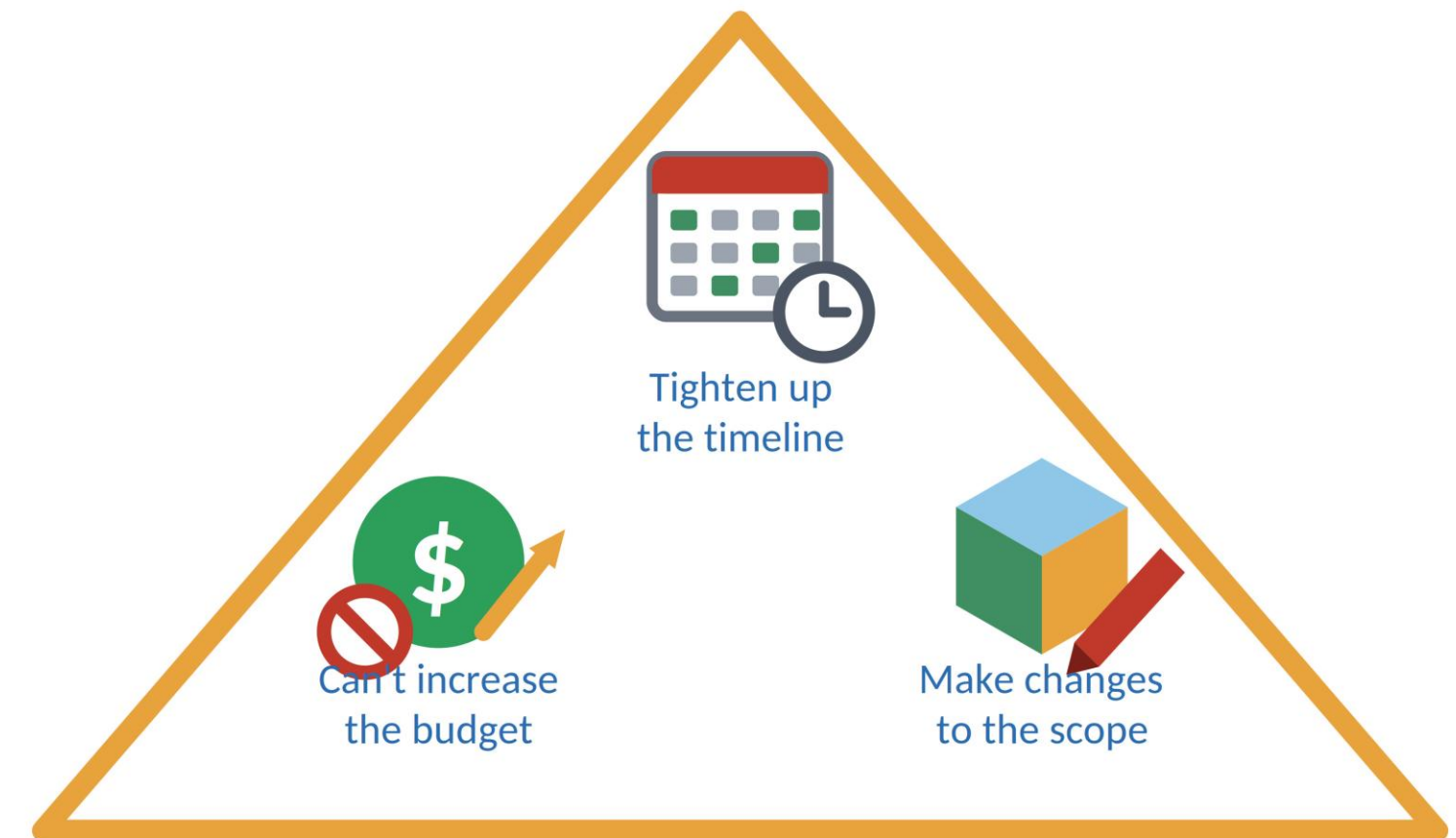
If out-of-scope work is required, be sure to document all costs incurred. That includes costs for work indirectly impacted by the increased scope. Be sure to indicate what the charges are for.

The script that works is not “no”, but “yes, and here is what it costs”.

Offer a trade-off



SCOPE Change — MOVE THE TIMELINE



TIMELINE FIXED — MOVE THE SCOPE

Keep one stakeholder happy, or deliver on time



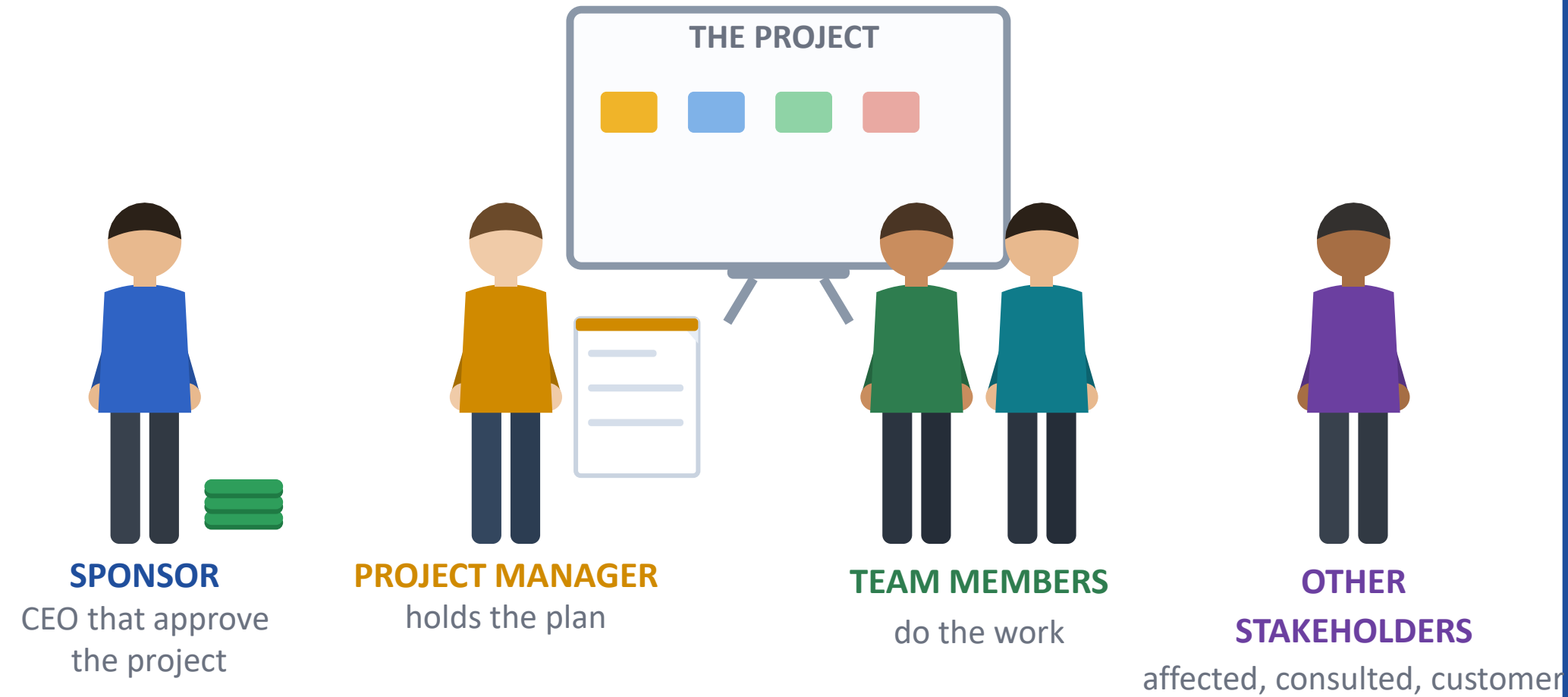
— PART 08

Stakeholders and roles

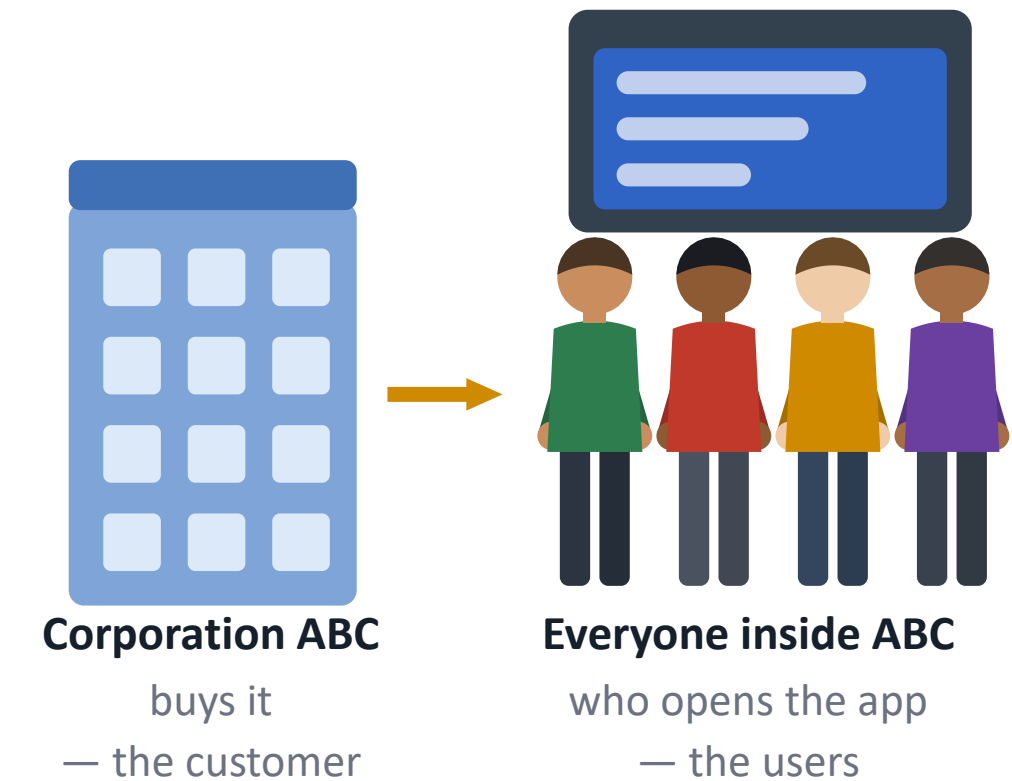
— PART 08

The roles on a project

PRIMARY STAKEHOLDERS



CUSTOMER OR USER?



— PART 08

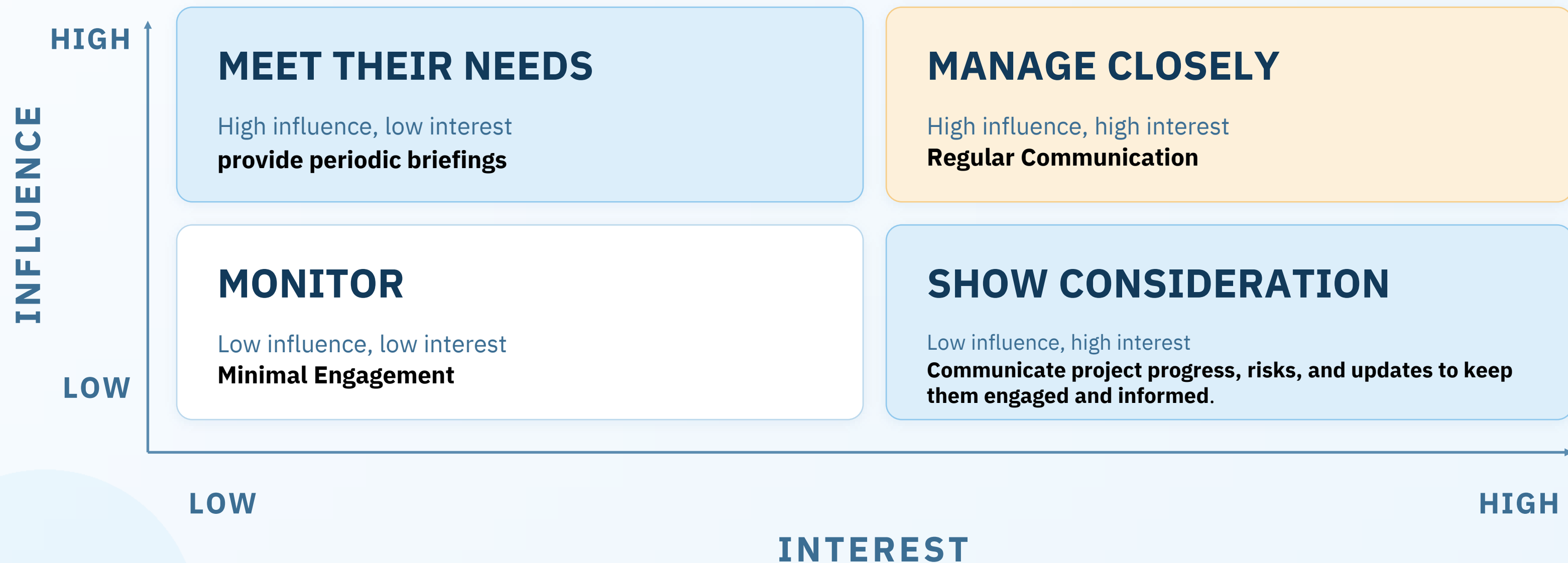
Stakeholder analysis

-  1 List everyone the project affects: who is invested, who is affected, who contributes
-  2 Determine the level of interest and influence of each — this identifies your key stakeholders
-  3 Assess each one's ability to participate, and find ways to involve them

The higher the interest and influence, the more their needs are prioritised.

— PART 08

The power grid



— PART 08 · WORKED EXAMPLE

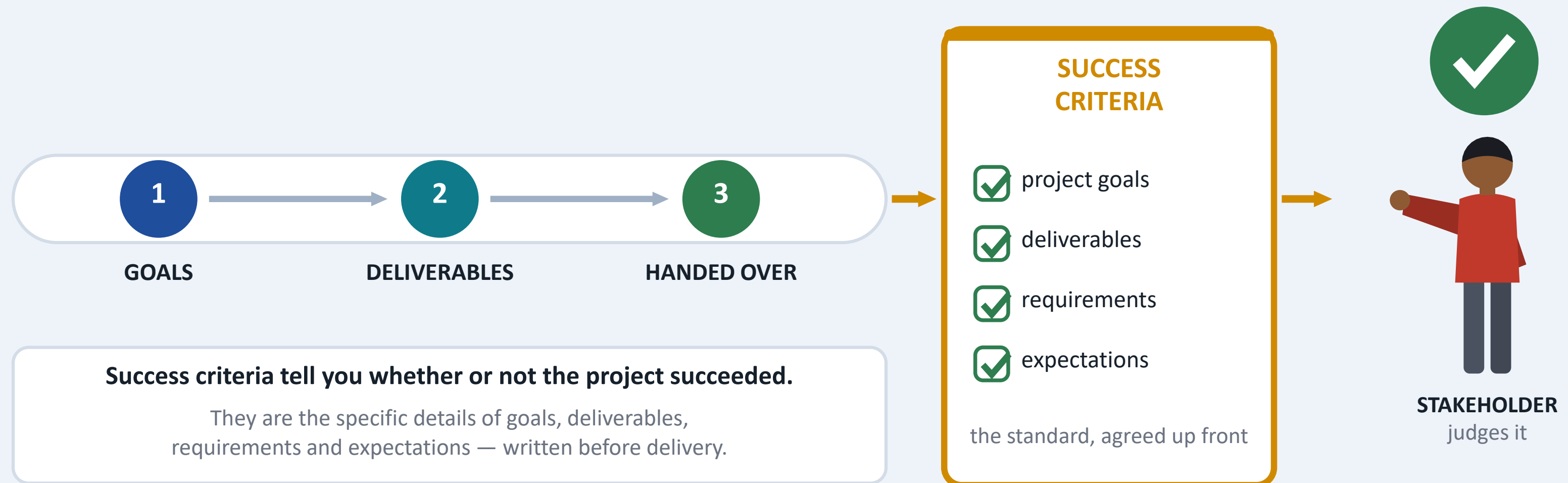
The power grid, filled in for TokTickIT



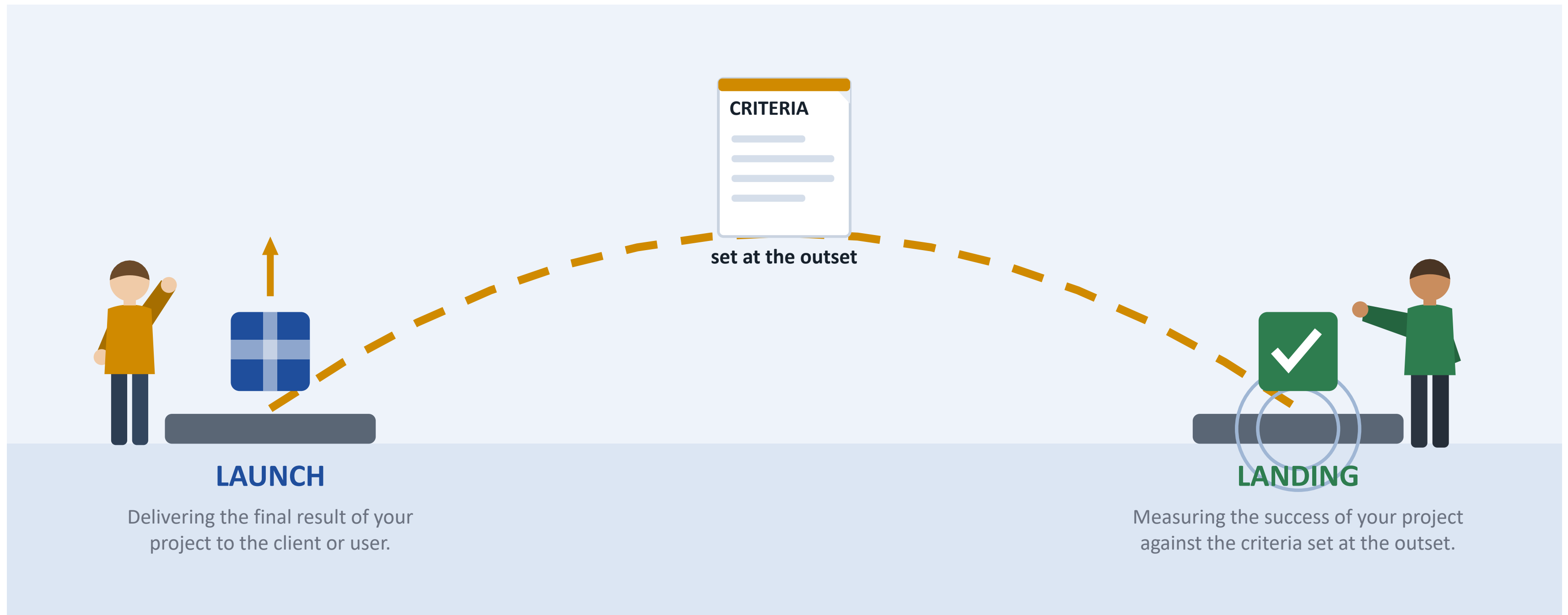
— PART 09

Success Criteria

Success criteria



Launch and landing

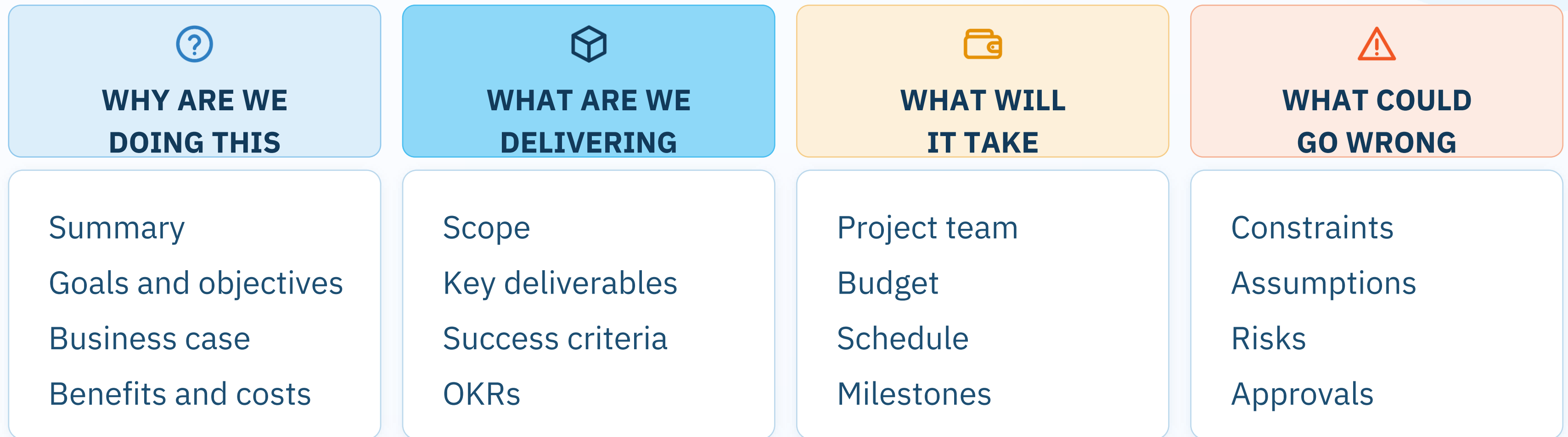


— PART 09

The project charter

— PART 09

What goes in the charter



Thirteen elements, and four questions that make them something other than a list to memorise.

— PART 09

There is no single charter format



Charters vary between organisations and between projects



A short, simplified charter suits a small, uncomplicated project



A complex project links to deeper analysis held in an appendix



Choosing the right form for the project is the project manager's call

It is a living document. Let it grow with the project, and revisit it often.

Project Charter — one page

Team

Sponsor

name

Project manager

name

Tech lead

name

Core team

name, name, name

Problem statement

What is wrong today, for whom, and what it is costing us.

Business case

Why it is worth solving now.

Success metrics

Measure	Target	Now
Cost saved	฿ X	—
Defects after release	X %	—
Time to deliver	X days	—
User satisfaction	X / 5	—

Timeline

Define the problem

Agree the target

Design the solution

Build and test

Hand over and sustain

Risks, deliverables and results

Top risks	Key deliverables	Key results
Risk 1	Deliverable 1	Result 1
Risk 2	Deliverable 2	Result 2
Risk 3	Deliverable 3	Result 3
Risk 4	Deliverable 4	Result 4
Risk 5	Deliverable 5	Result 5

The charter template, page by page

1 • Summary, goals, deliverables, scope

Project Charter Template

Project Summary

Goals

The desired outcome of the project (clear, specific and measurable)

Specific

- What do I want to accomplish?
- Why is this a goal?
- Target group?
- Where should the goal be delivered
- To what degree?

Measurable

- How much?
- How many?
- How will I know when it's accomplished?

Attainable

- Can goal be reached?
- How can it be accomplished?

Relevant

- Does the goal make sense?
- is the goal worth while?
- is the right time?

Time-bound

- When is the deadline?

Deliverables

The products or services that are created for the customer, clients, or project sponsor

Scope

Charter template redrawn for CPE 334 — layout and artwork original; field wording unchanged.

2 • In and out of scope, benefits and cost

In Scope:

Out of Scope:

Benefits & Cost

Benefit: (answer all that apply)

Benefits	gain
What value will this project create?	
How much money could this project save our organization?	
How much money will it bring in from existing customers?	
How much time will it save?	
How will it improve the customer experience?	
SUM	

Intangible Benefit: (answer all that apply)

Benefits	gain
Will the project increase customer?	
Is the project likely to improve employee morale?	
Will the project reduce employee's hours?	
Is the project likely to improve the company's brand perception and recognition, attracting more customers or providing a competitive advantage?	
SUM	

Cost: (answer all that apply)

Costs	value
How much time will people have to spend on this project?	
What are the one-time costs?	
Are there any ongoing costs?	
What about long-term costs?	

3 • Risk, ROI, success criteria, stakeholders, resources

Risk Factor, what can go wrong that cost us money, use expected value. Example: there are 10% chance this problem will happen that cost company 100,000 baht = 0.1 x 100,000 = 10000

SUM

Intangible Cost: (answer all that apply)

Costs	value
Will the project reduce customer?	
Is the project likely to reduce employee morale, sanctification?	
Is the project likely to reduce the company's brand perception and recognition?	
Will taking this project prevent us from getting other project?	
SUM	

ROI = (SUM of Benefit - SUM of costs) / Sum of costs (no intangible will be used for ROI)

ROI:

Success Criteria

Stakeholders

Resources

Charter template redrawn for CPE 334 — layout and artwork original; field wording unchanged.

81

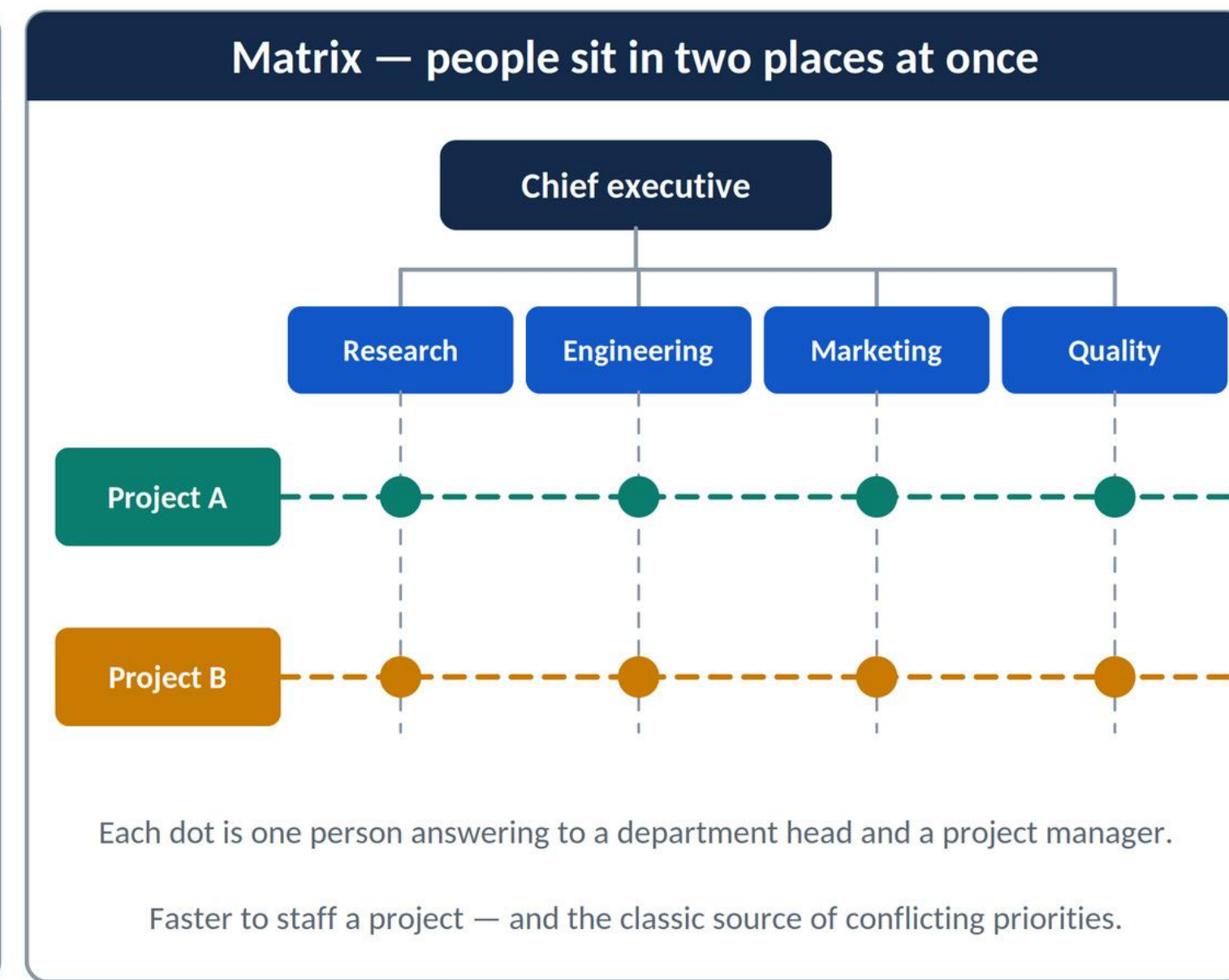
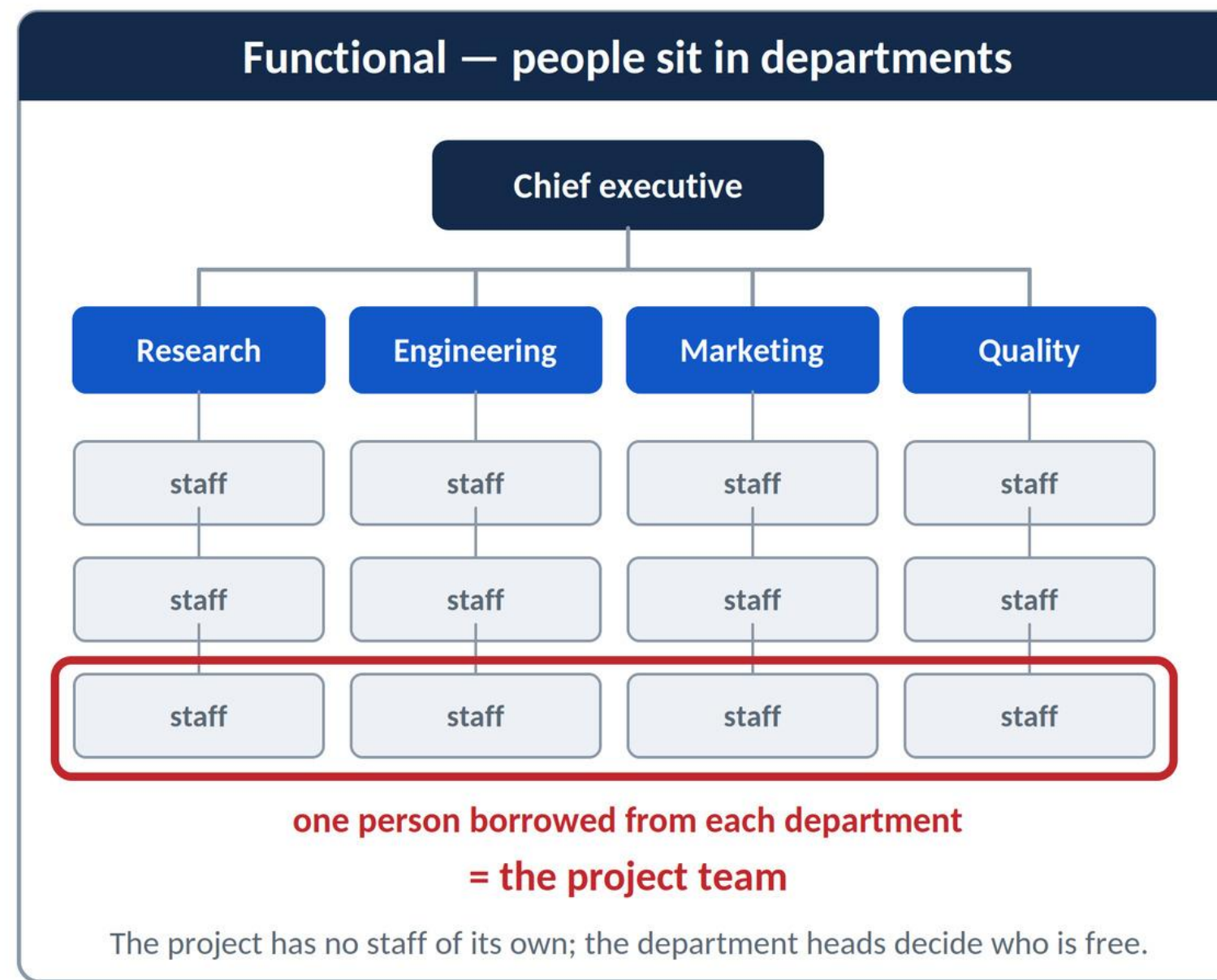
— PART 02

Project Planning

— STRUCTURE DECIDES AUTHORITY

Organization structure

Where the Project Team Comes From



Which structure you are in decides how much authority a project manager actually has.

— CLASSIC VS MATRIX

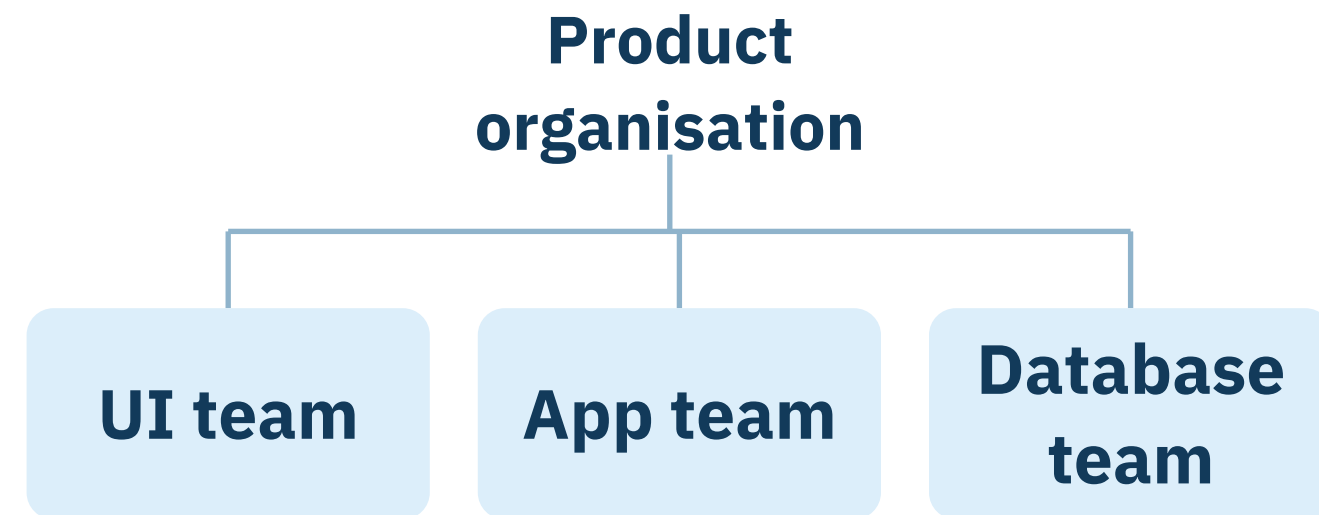
The two impacts

	TRADITIONAL	MATRIX
REPORTING	One manager	Two or more chains of command
THE PM'S AUTHORITY	Limited, and borrowed	More autonomy to decide
GETTING RESOURCES	Request up the chain	Negotiate and share
CHAIN OF COMMAND	Clearly defined	Not always defined
THE RISK	Slow approvals	Divided loyalty, competing priorities
BOTH	The PM must define roles and responsibilities explicitly.	

— CONWAY'S LAW, 1968

You ship your org chart

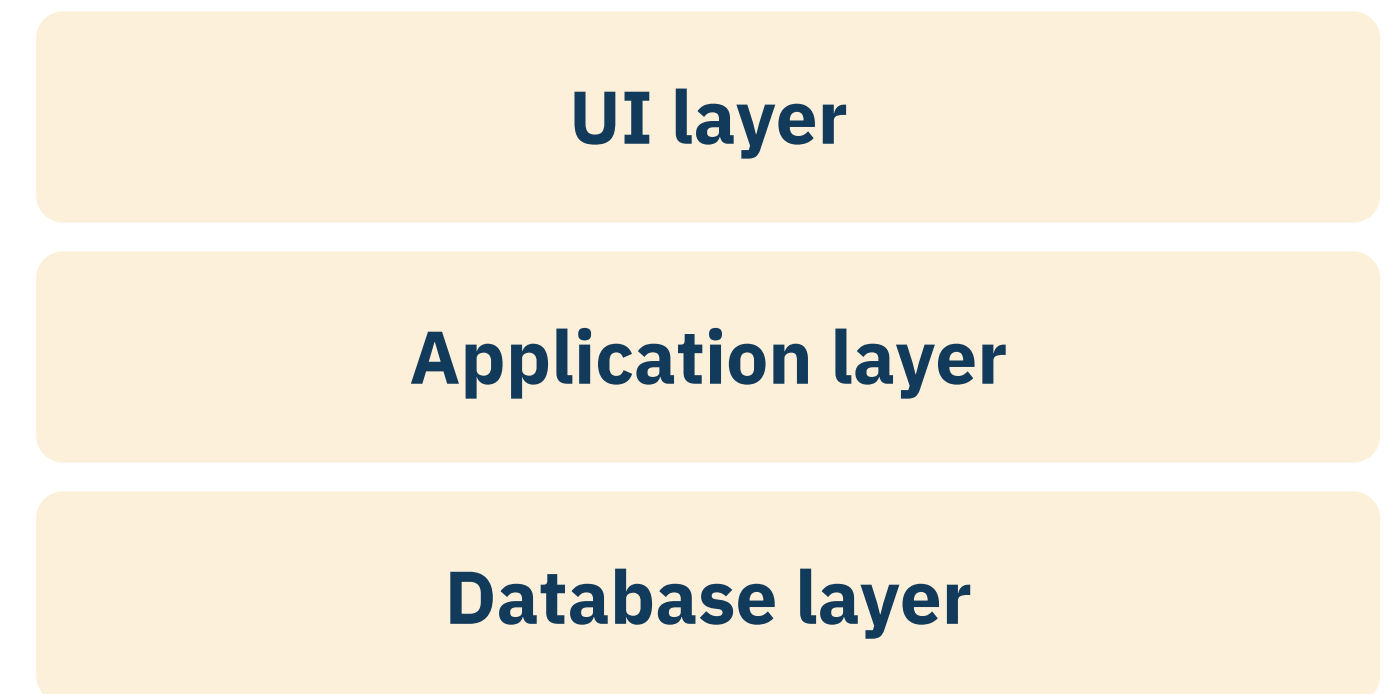
THE ORGANISATION



PRODUCES



THE SYSTEM SHIPS



Organise teams around business areas instead — loosely coupled, tightly aligned, each owning its area end to end.

— THE WALL OF CONFUSION -Andrew Clay Schafer

Two teams, opposite measures

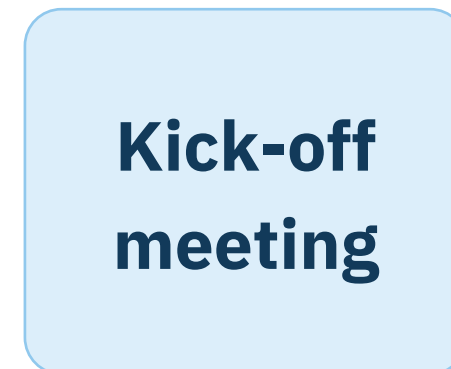
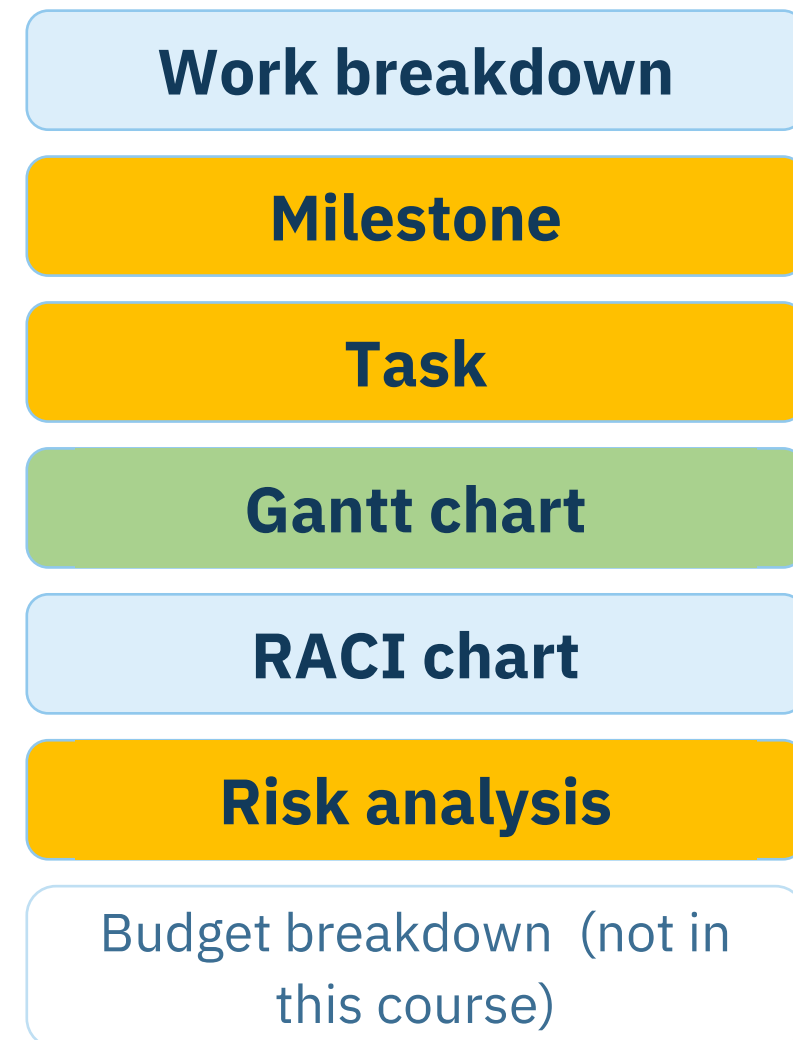


Sprinting from January. Deployed in September.

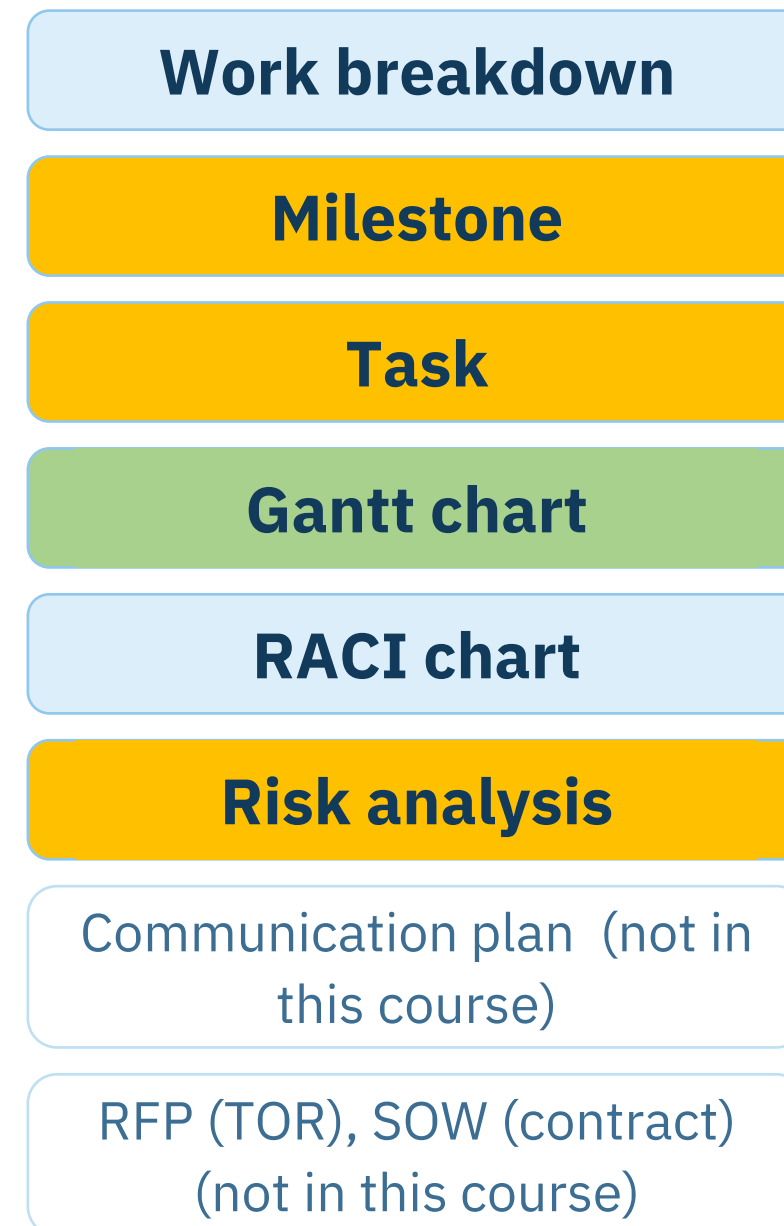
No amount of PM (Agile) on one side fixes the other.

What a project plan is made of

DRAFT THIS



REFINE THIS

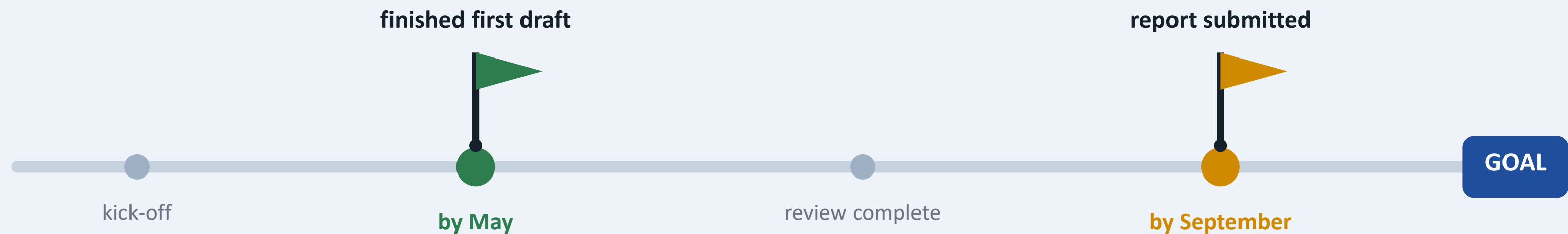


PROJECT PLAN



Vendor and procurement work only applies if your project buys something in.

Milestone



Milestone

An important point within the project schedule that indicates progress and usually signifies the completion of a deliverable or phase of the project.

GOAL vs MILESTONE

Goal → deliver the technical report

Milestone → the first draft is finished

A milestone is a point in time, not a piece of work.

Benefits of setting milestones



Project task

Project task

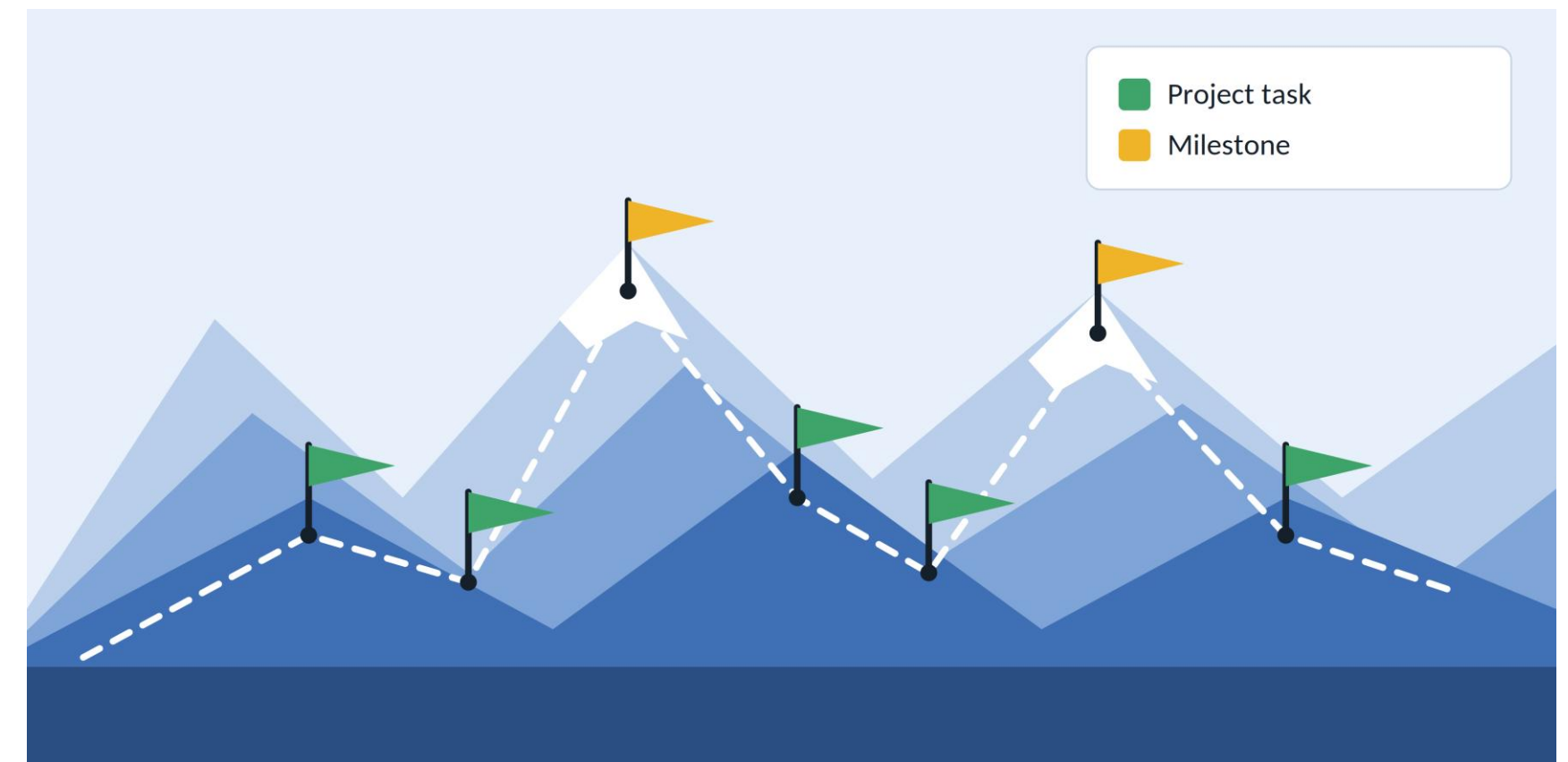
An activity that needs to be accomplished within a set period of time.

FROM MILESTONE TO TASKS

Milestone → finished first draft

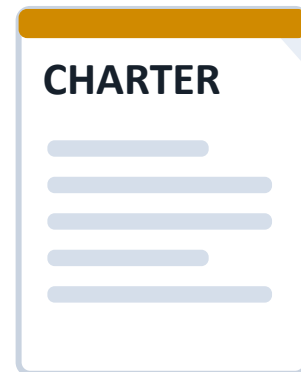
Tasks →

- hiring a writer
- conducting research
- drafting the different sections of the report



How to set a milestone

1 Evaluate the whole project



The project charter already lists the deliverables and the phases — start there.

2 Apply the review test



3 Put them in sequence

MILESTONES OF A WEB PROJECT



If the stakeholders would not need to review it,
it is (mostly) a task — not a milestone.

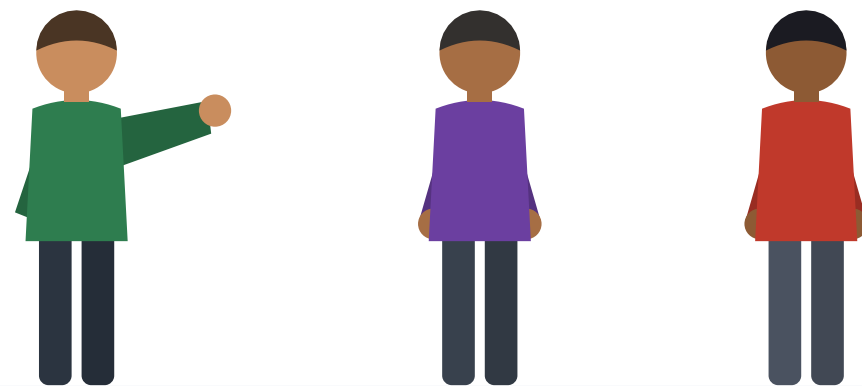
Giving each milestone a deadline

ASK THE PEOPLE WHO WILL DO IT

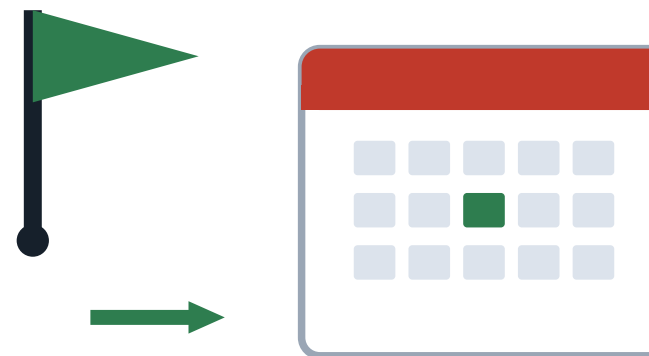
TASKS TO REACH IT



“about three days”



EACH MILESTONE GETS A DATE



milestone + deadline

- 1 Estimate the tasks that reach it
- 2 Add them up, with the team
- 3 Then commit to the date

AND THE STAKEHOLDERS' NEEDS



“The board meets on the 12th.”

When determining deadlines for milestones, consider the needs of your stakeholders.

Set tasks to identify milestones

TOP-DOWN

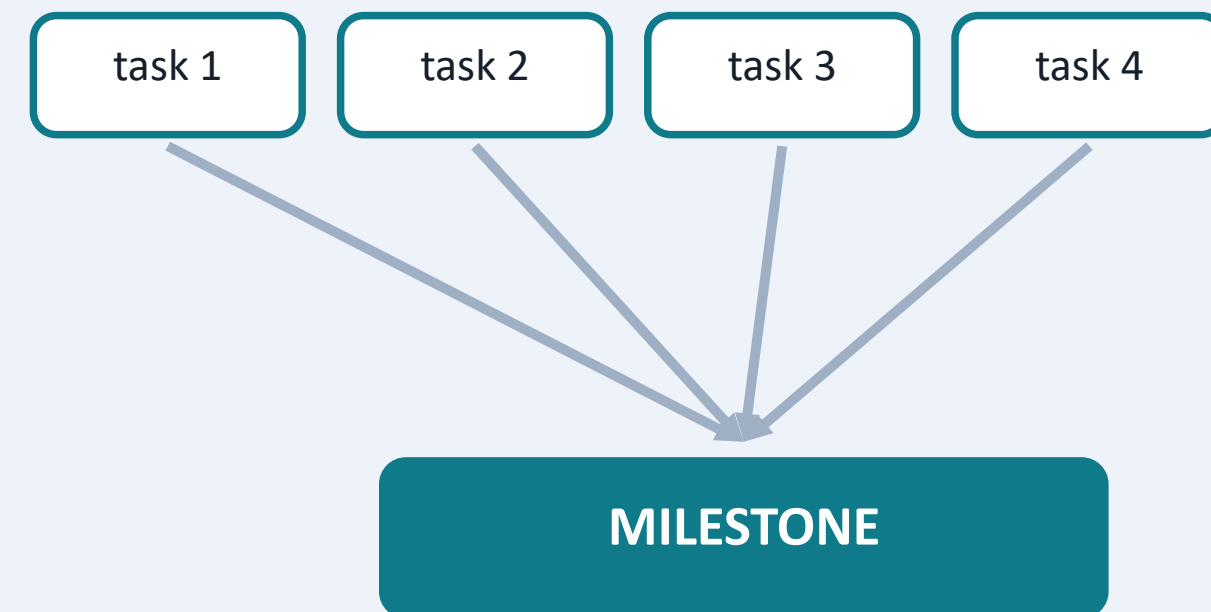
milestones first, then break them down



The PM lays out the milestones, then works with the team to catch every task.

BOTTOM-UP

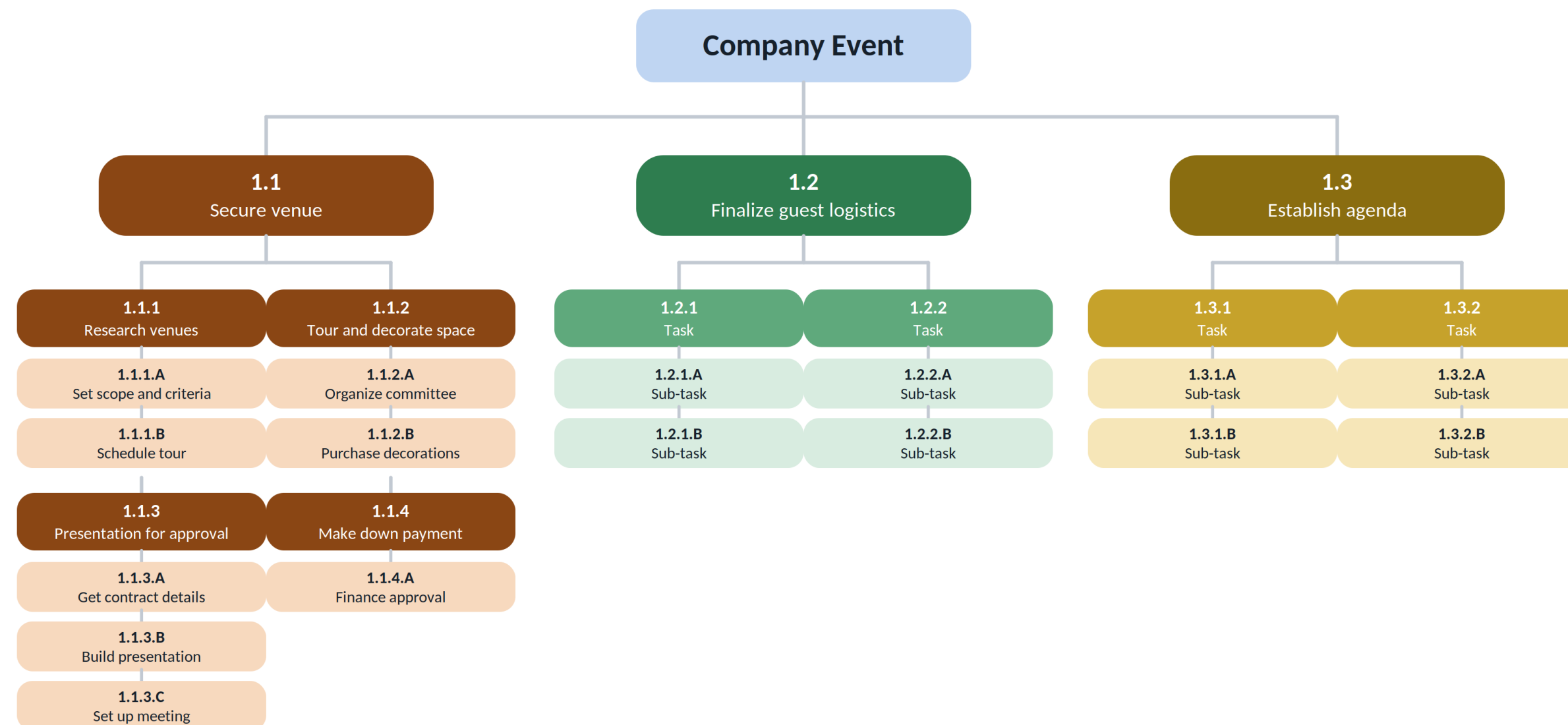
tasks first, then roll them up



The PM looks at every individual task, then rolls them into chunks that reach a milestone.

Work breakdown structure (WBS)

A tool that sorts the milestones and tasks of a project in a hierarchy, in the order they need to be completed.



— THREE STEPS

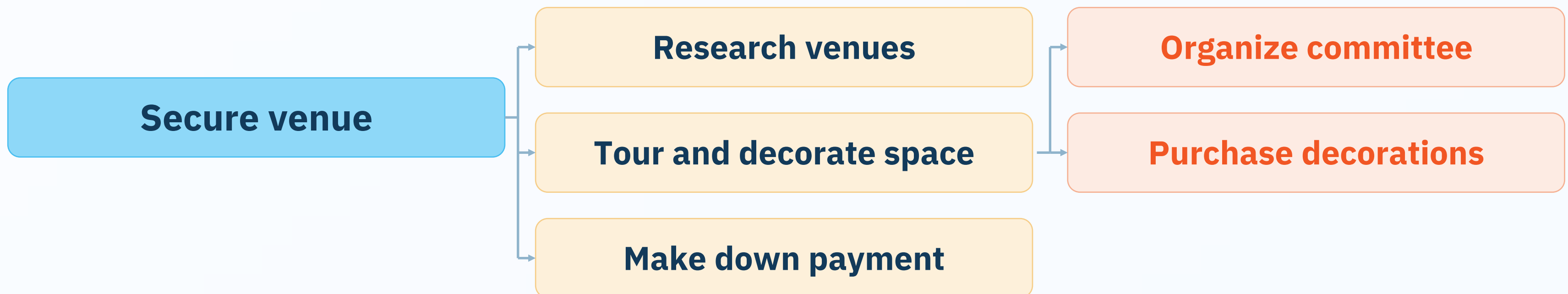
Building one, from the top down

1 List the milestones

2 Find the tasks that meet each milestone

3 Break those tasks into sub-tasks

A WORKED BREAKDOWN • PLANNING A COMPANY EVENT



— TOKTICKIT

The breakdown for Lab 3, week 2

```
|— A Authorization
|   |— T1 Role guards and ownership checks
|— B IT Staff ticket queue
|   |— T2 Queue API: search, filter, sort, page
|— C IT Staff ticket detail
|   |— T3 Claim / reassign, IT Priority, status
|— D Comments and notes
|   |— T4 Public Comments, Internal Notes
|— E Administrator user management
|   |— T5 User list, create, edit, activate
|— F Verification
|   |— T6 Authorization, regression and E2E tests
```

Capacity planning and the critical path

Capacity planning

The act of allocating people and resources to project tasks, and determining whether or not you have the necessary resources required to complete the work on time.

Critical path

The list of project milestones you must reach in order to meet the project goal on schedule, as well as the mandatory tasks that contribute to the completion of each milestone.

Your critical path includes the bare minimum number of tasks and milestones you need to reach your project goal.



Mock up
designs



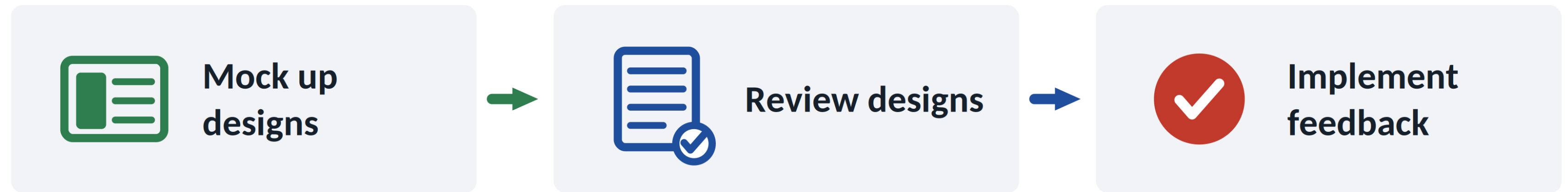
Review designs



Implement
feedback

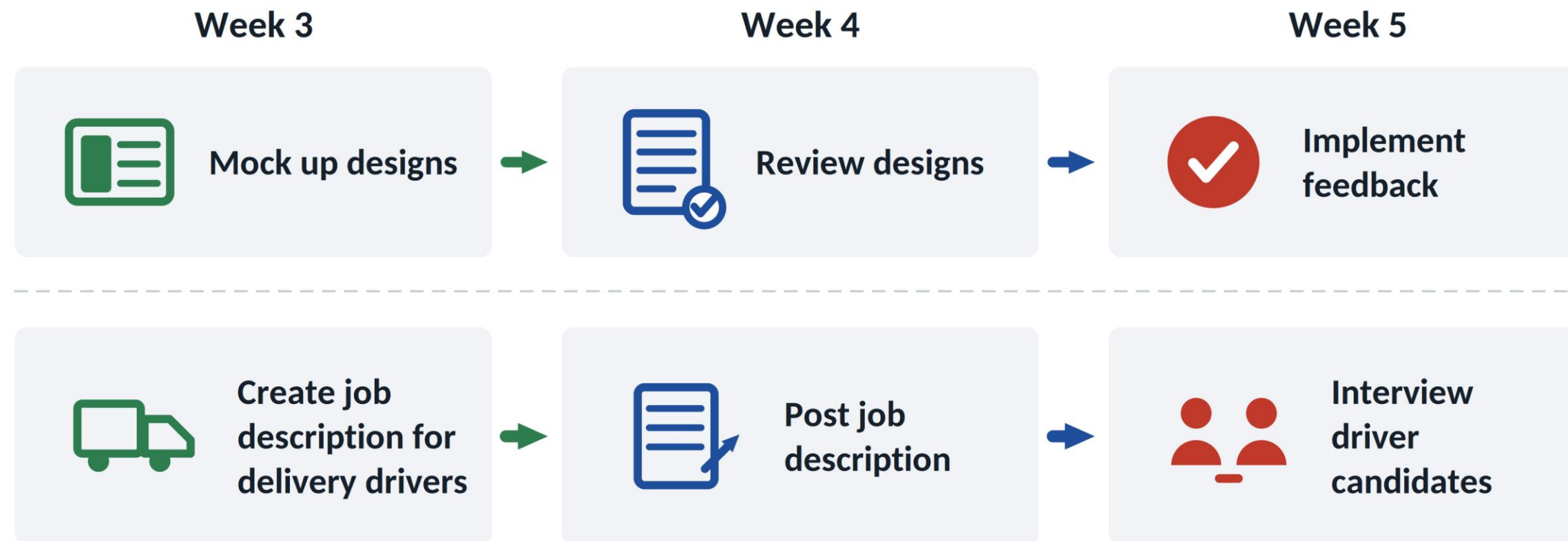
Dependency and the critical path

DEPENDENCY



Start to finish, the longest path through the network is the critical path.

Parallel or sequential?

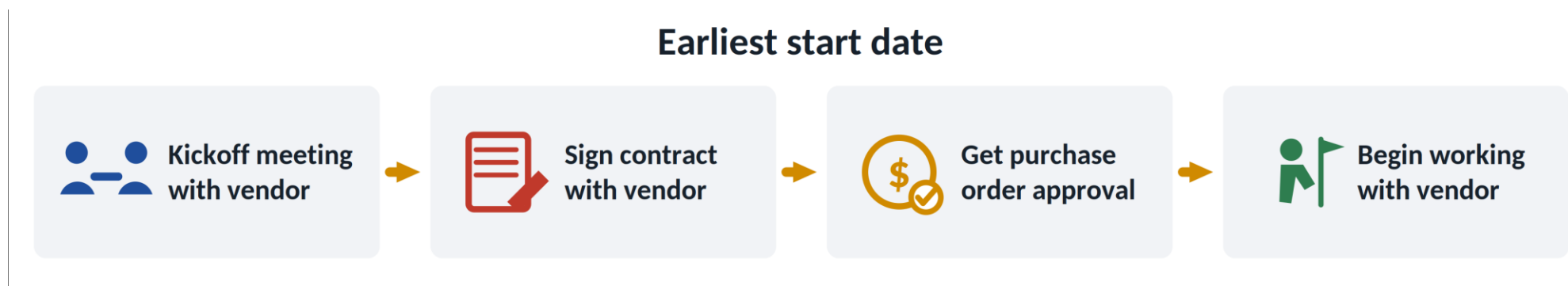


Identify which tasks can happen in parallel, and which must happen sequentially

Determine which project tasks have a fixed start date

Earliest start date and float

EARLIEST START DATE



Determine which project tasks have an earliest start date

Identify if a task has float (also called slack)

Float is the amount of time you can wait to begin a task before it impacts the project schedule and threatens the project outcome.

Tasks on the critical path should have zero float.

— TWO PASSES

Forward, then backward

FORWARD PASS

The earliest each task can start and finish.

START

DEADLINE

The latest each one can start without moving the deadline.

BACKWARD PASS

Time and effort estimation

The same task, measured two different ways

TIME ESTIMATION

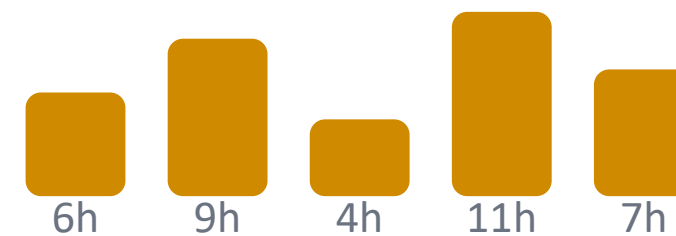
how long it will take on the calendar



A prediction of the total amount of time required to complete a task.

EFFORT ESTIMATION

how much work, and how hard



A prediction of the amount and difficulty of active work required to complete a task.

— THE PLANNING FALLACY

Why the estimate is always short

We underestimate time, cost and risk — we picture the run where nothing goes wrong.

THE PLAN

Walk the dog · 20
min

WHAT HAPPENS

The walk

Rain

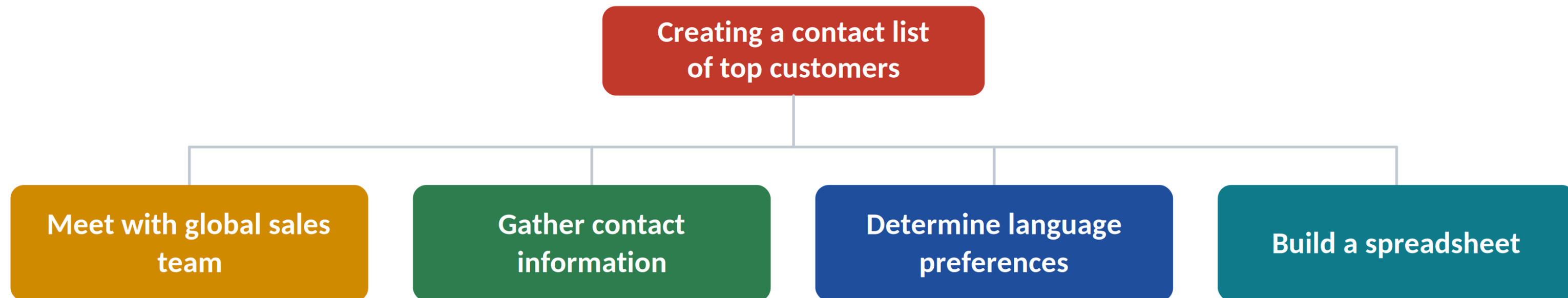
Another dog

Sniffing

Kahneman and Tversky, 1977

Buffer

Extra time added to the end of a task or project to account for unexpected slowdowns or delays in work progress.

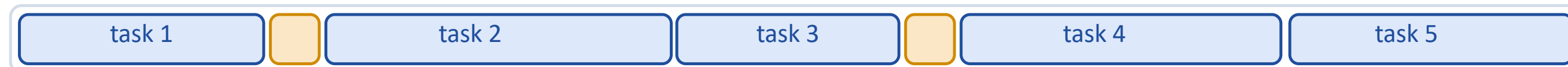


Types of scheduling buffer

Where the slack sits

TASK BUFFER

extra time on the end of a single task



PROJECT BUFFER

extra time on the end of the whole project



one pool, held at the end, spent only when needed

Use a task buffer where the delay is outside your team's control. Otherwise, hold one pool at the end.

— METHOD

Three-point estimation

$E = (O + 4M + P) / 6$ # expected hours

$SD = (P - O) / 6$ # uncertainty

O optimistic everything works on the first try

M most likely your honest guess

P pessimistic the day the migration breaks

PERT formula from Navy Polaris project 1958

Estimating the six tasks

	Task	O	M	P	E
T1	Role guards and ownership checks	2	3	6	3.33
T2	Queue API: search, filter, sort, page	3	5	10	5.50
T3	Claim / reassign, IT Priority, status	4	6	11	6.50
T4	Public Comments and Internal Notes	4	7	13	7.50
T5	Admin users: list, create, edit, activate	3	5	12	5.83
T6	Authorization, regression and E2E tests	4	6	14	7.00
	TOTAL	20	32	66	35.67

— THE ANSWER

**Report a range,
not a number**

35.7 h $\pm$ 3.2 h

- ◆ **Expected** — 35.7 hours of effort
- ◆ **One sigma** — 32.5 to 38.9 hours
- ◆ **Plan against** — 42 hours — two sigma

— WITH AN AGENT

The hours move, they do not vanish

Typing the code

DOWN

Reviewing the diff

UP

Writing the tests

Down

Integrating and debugging

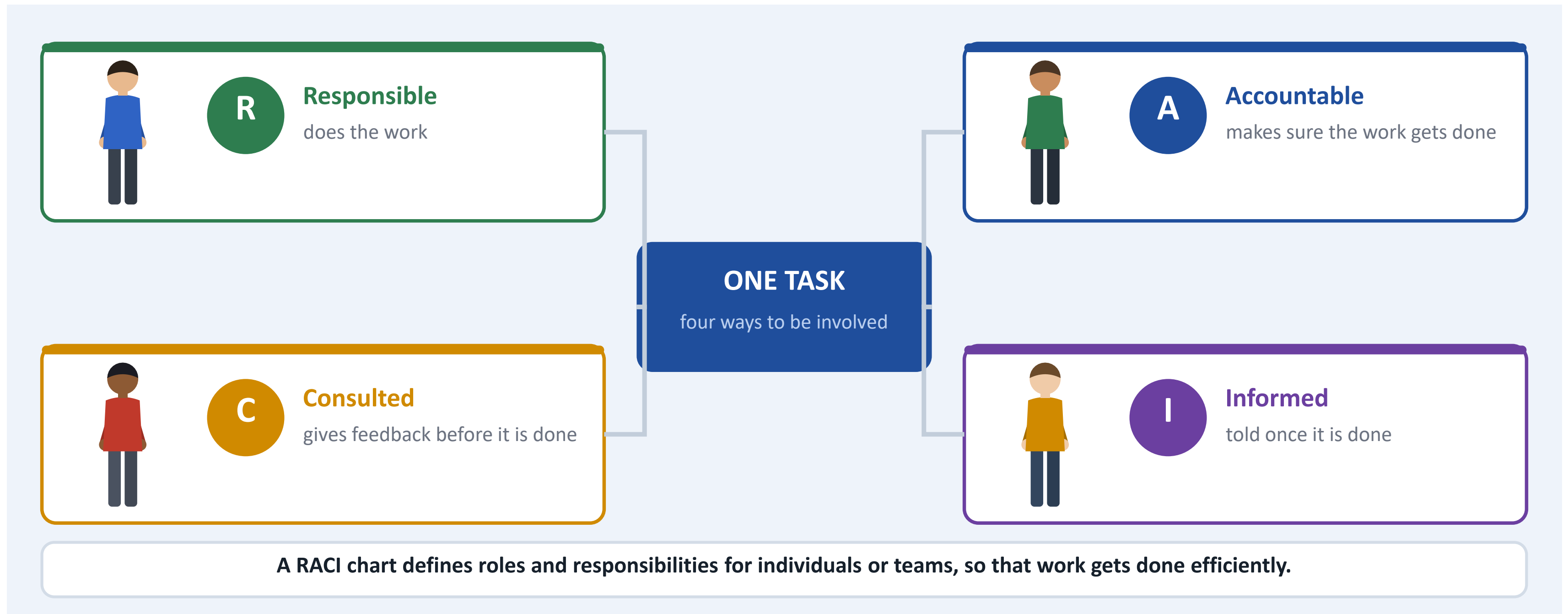
UP

Project Schedule — Gantt Chart



Each bar shows when a task starts, how long it runs, and who owns it.

RACI



Reading a RACI chart

Step	Project initiation	Project executive	Project manager	Project analyst	Project architect	Application developers
1	Task 1	C	A/R	C	I	I
2	Task 2	A	I	R	C	I
3	Task 3	A	I	R	C	I
4	Task 4	C	A	R	R	I

R responsible • A accountable • C consulted • I informed

WATCH THE WORKLOAD

Are there too many tasks on one person? Tally the Rs against each stakeholder — that is how you spot someone being overloaded.

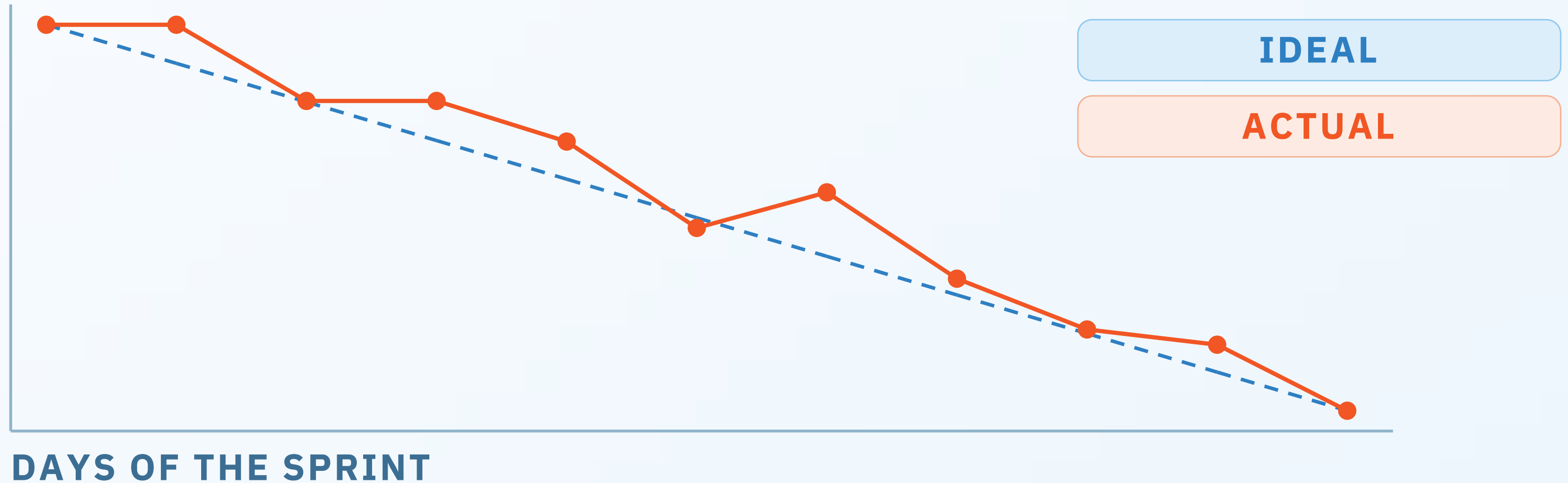
WHEN IT IS WORTH IT

Larger projects benefit greatly from one
On a very small team, with few stakeholders, clear roles and a short timeline, a RACI chart could actually slow the project down.

— THE BURNDOWN

Story points remaining against time

POINTS
LEFT



Budget planning

- 1

Break the project into tasks
- 2

Estimate the cost of each item
- 3

Add the estimates together
- 4

Add contingency and tax
- 5

Seek approval from key stakeholders

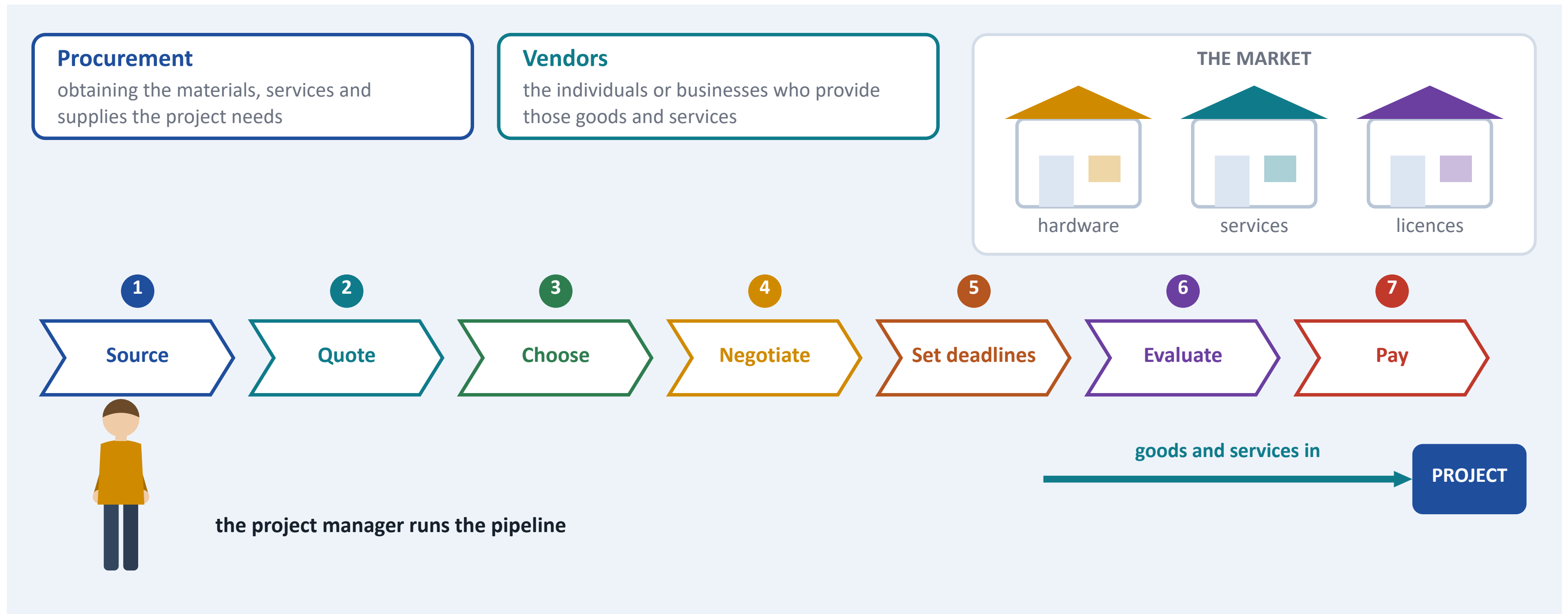
					TARGET BUDGET	ACTUAL BUDGET	UNDER/OVER	
					\$	\$	\$	
	Labor		Materials		Fixed cost	Budget	Actual	Under/Over
TASK	Hrs	Rate	Units	\$/Unit				
CATEGORY								
Task					\$			\$
Task					\$			\$
Task					\$			\$
Task					\$			\$
					\$	\$		
CATEGORY								
Task					\$			\$
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CATEGORY								
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Task					\$			\$
Task					\$			\$
					\$	\$		
CATEGORY								
Task					\$			\$
Task					\$			\$
Task					\$			\$
TOTAL					\$			\$

Budget template redrawn for CPE 334 — one row per task, rolled up per category.

The reward for finishing below budget



Procurement and vendors



Agile and traditional procurement

AGILE PROCUREMENT

collaborative, with a living contract



- team and supplier work together
- emphasis on the relationship between the parties
- the team helps decide what to buy

TRADITIONAL PROCUREMENT

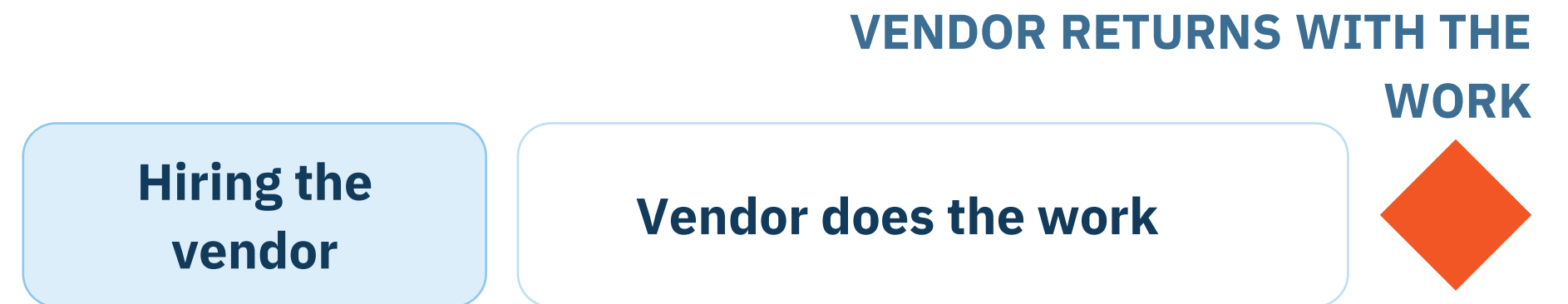
standard contracts with clear terms



- standard contract, clear deliverables
- the PM may own procurement end to end
- documentation is lengthy and extensive

Hiring a vendor: expectation vs reality

EXPECTATION



REALITY



The vendor is a huge source of risk — the check-ins are what keep it visible.

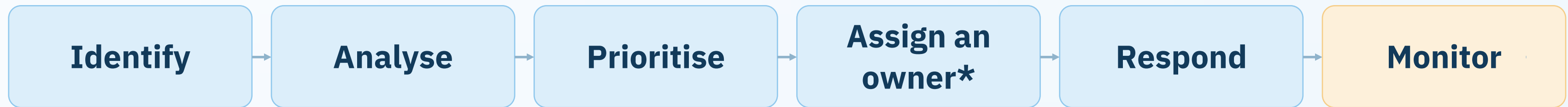
Risk management plan

A living document that contains information on the high-level risks and the mitigation plans for those risks.

ID	Risk Description	Risk to Project	Mitigation Plan
1	Vendor is one day behind schedule to meet an important milestone	Low	Check in with the vendor to determine if the project manager can help remove any blockers to progress
1.1	Vendor is far behind schedule to meet an important milestone	Medium	Hold daily check-in meetings to get updates on progress and remove blockers where necessary
1.2	Vendor misses an important deadline	High	Hire a new vendor

— PLANNING · RISKSIX STEPS

The risk management process



The risk management plan

The risk	Probability	Impact	Owner	Planned response
Staging environment breaks	Medium	High	Nun	Docker compose file in the repo, tested weekly
Teammate unavailable in week 2	Low	High	Ploy	Pair on T2 so two people know it

Types of risk, impact and mitigation

TYPES OF RISK

- Time risk
- Budget risk
- Scope risk
- External risk — environmental, legal
- Single point of failure
- Dependency — internal and external

FOUR TYPES OF RISK MITIGATION

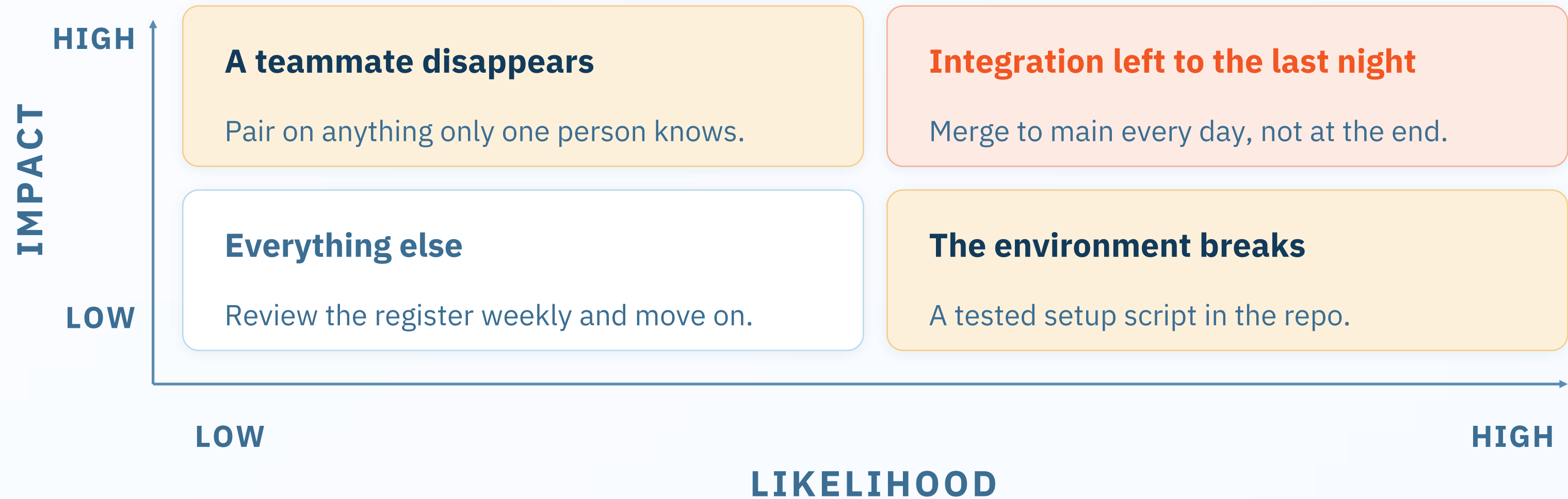
Minimize example: Multi-factor authentication reducing the likelihood of credential-based cyber incidents



Inherent Risk				
		Impact		
		Low	Medium	High
Probability	High	Medium	High	High
	Medium	Low	Medium	High
	Low	Low	Low	Medium

— ON YOUR OWN PROJECT

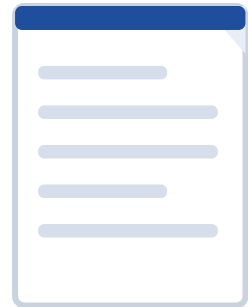
The three that hit student teams every year



Communication plan

Type of Communication	Recipients	Frequency	Key Dates	Delivery Method	Goal	Sender/Owner
Project Newsletter	Key Stakeholders	Monthly	First Monday	Email	Status Update Overview	Project Manager
Daily Stand-Ups	Core Team	Daily	12pm ET	In Person / Video Conference	Progress Updates, Blockers, Next Steps	Team Lead
Weekly Check In	Marketing	Weekly	Wednesdays 2pm	Email + Video Conference		Project Manager
Weekly Check In	Procurement	Weekly	Wednesdays 3pm	Email + Video Conference		Project Manager / Team Lead
Weekly Check In	Product Development	Weekly	Wednesdays 4pm	Email + Video Conference		Project Manager

Quick reference — the project documents



Project charter

why we are doing this, and who agreed



GANTT chart

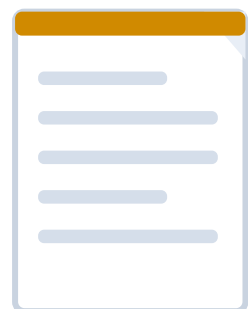
what happens when, and who owns it



RACI chart

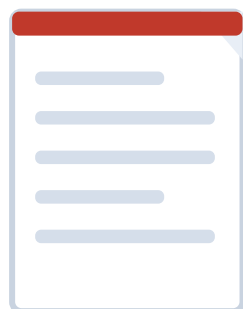
optional

who is R, A, C and I on each task



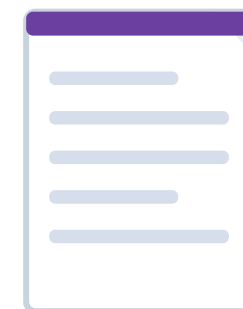
Risk management plan

the risks we know about



Budget and Procurement

Where the resource and help come from



Communication plan

who hears what, how often

THE KICK-OFF

**The first meeting of the team,
the stakeholders and the sponsor.**

At the start of a project, or of a new phase.

— THE KICK-OFF

Project kick-off meeting

The first meeting in which a project team comes together to ground everyone in a shared vision, gain a shared understanding of the project's goals and scope, and understand each person's individual role within the team.

WHO TO INVITE

Project team

Project stakeholders

Project sponsor

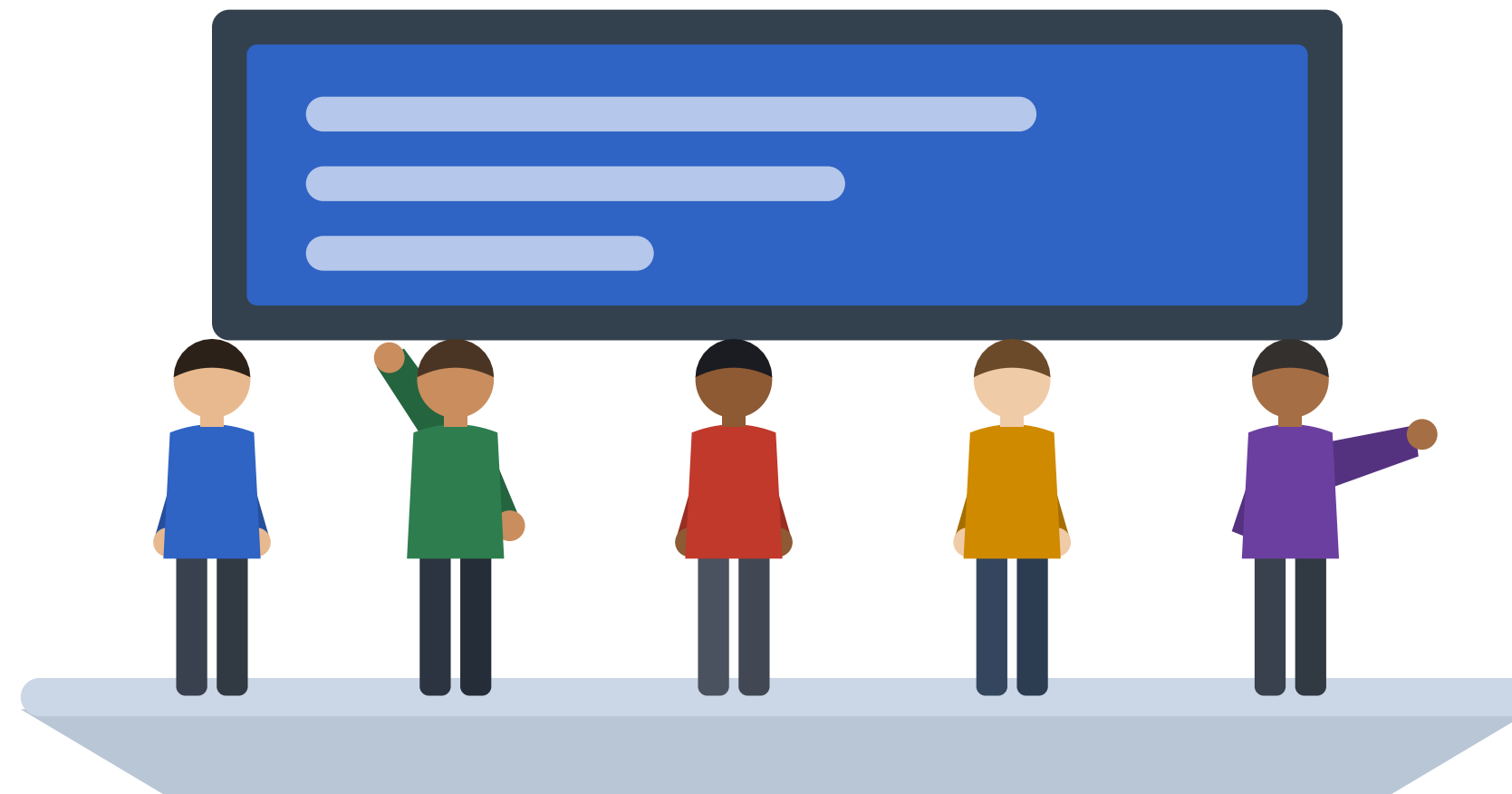
— WHY IT MATTERS

What the kick-off has to achieve

Establish a shared vision

Align on scope

Build team rapport



Ask questions, offer insights

Set expectations

— THE AGENDA

Seven items in one hour

10 min

Introductions

- Team member names
- Project roles

5 min

Background and purpose

- How the project came to be
- Why the project matters
- Set a shared vision

5 min

Goals and scope

- In-scope
- Out-of-scope
- Target launch date
- Milestones

5 min

Roles and responsibilities

- What work everyone is responsible for throughout the project

10 min

Collaboration process and tools

- Shared project tools and documents
- Communication expectations

10 min

What comes next

- Set expectations and action items

15 min

Questions

- Gain clarity on meeting topics
- Ensure the project benefits from diversity of thought, experience and ideas

0 min

60 min
121

— DURING AND AFTER

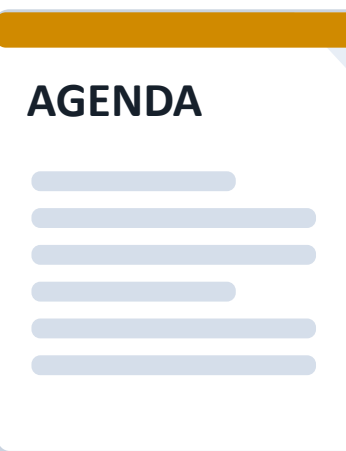
Best Practices

DURING



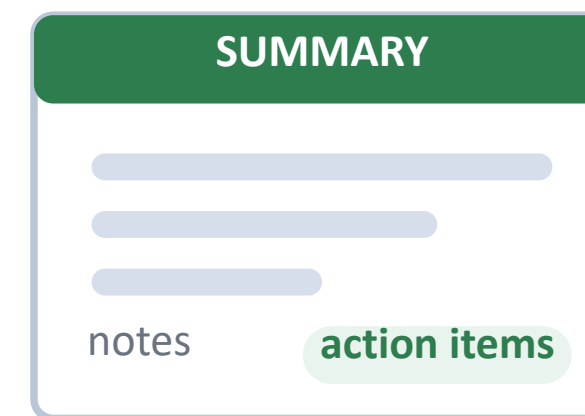
NOTETAKER

designated before you start



keep redirecting
to the agenda

AFTER



sent to the whole team



“

Adding manpower to a late software project makes it later.

— Fred Brooks, The Mythical Man-Month, 1975

— LAB 3

What you build, and what you hand in

BUILD THIS SPRINT

- **Role-based authorization**
Requester · IT Staff · Administrator, enforced on the server
- **IT Staff Ticket Queue**
search, filters, sorting, pagination
- **IT Staff Ticket Detail**
claim / reassign, IT Priority, permitted status
- **Public Comments and Internal Notes**
append-only; notes hidden from Requesters
- **Administrator User Management**
list, create, edit, one role, activate or deactivate

NOT THIS SPRINT

Actions Taken

Lab 4

Dashboards and KPIs

Lab 4

Email, SSO, self-registration

not in this course

Cloud deployment

Week 11

HAND IN

```
docs/lab-03/specification.md
docs/lab-03/tests.md
docs/lab-03/ui-spec.md
docs/lab-03/api-spec.md
docs/lab-03/reviewer.md
docs/lab-03/ai-use.md
```

feature branch → lab3-staging → main

one PDF, headed Answer Part 1 – 9

the final main branch is the source of truth

10

P1

Git and workflow

5

P2

Spec DD

10

P3

Test DD

5

P4

AI use

5

P5

Login UI

5

P6

Queue UI

10

P7

Ticket detail UI

5

P8

Admin users
UI

5

P9

UI and
responsive

Questions?

Next: the plan becomes a board, and the board becomes a sprint.

